Inferential targets for reproduction-number estimates: counting error, between-period variation and what cross-sectional surveillance diagnostics measure

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Abstract

Case counts are often taken to set the precision of reproduction-number estimates, yet the uncertainty that additional cases remove depends on the inferential target. In a one-generation branching model with negative binomial offspring and binomial reporting, we derive exact mean squared errors for three targets: the current reproduction number, the realised mean over observed periods and the process mean. Counting error falls inversely with the expected count for every target, whereas the process mean also retains between-period transmission variance. Dividing by the previous observed count instead of a known scale raises the counting error by about half under illustrative parameters. When the reporting probability fluctuates between periods, single-period count means and variances cannot separate reporting volatility from transmission variation. We then audit cross-sectional dispersion diagnostics against known truth on the design of 1,748 state-weeks of US influenza hospital admissions (2022–2026). The intercept of an affine fit of relative squared deviation on inverse count is a count-independent relative dispersion: transmission variation under fixed reporting, and its sum with reporting shocks otherwise. In the admissions data it was 0.0150 (95% confidence interval 0.0101–0.0206). The pooled counting slope was 1.26 (0.81–1.73) times the independent-Poisson reference, and its ratio to a branching-Poisson reference depends on the unknown reporting probability. The dispersion-to-counting crossover count was 37 admissions per week (18–72); tests of equal season intercepts did not reject ($p = 0.14$ and 0.21). Counting slopes and thinning responses do not distinguish overdispersion, under-reporting and heterogeneous transmission; that requires temporal structure or external reporting data.

Keywords: reproduction number; inferential target; surveillance; overdispersion; reporting variation; empirical Bayes

1 Introduction

Reproduction numbers summarise transmission under specified conditions and are routinely estimated from surveillance counts [1, 2]. Renewal methods relate incidence to past infectiousness through a generation-interval distribution, and the width of their credible intervals shrinks as the number of cases grows [3, 4]. Work on low incidence, reporting noise and timely detection has made the limits of period-specific inference explicit [5–7], and temporal aggregation and reporting delays separate the reproduction number of the transmission process from the apparent reproduction number of an observed series [8–10]. A common reading of these results is that case volume sets the precision of the estimate. That reading leaves the target unspecified. The current transmission level, the average of transmission levels over the observed periods and the mean of the process that generates successive periods are different quantities, and variation between periods is signal for the first but sampling variation for the third.

The variance structure behind this distinction is classical. Variance-component models split the variance of a unit-level estimate into a within-unit sampling term and a between-unit term [11, 12]; random-effects meta-analysis estimates the between-study component on top of within-study error [13]; and small-area estimation and empirical-Bayes disease mapping weight each direct estimate against a common mean according to the ratio of the two terms [14–16]. Funnel plots for institutional comparison draw control limits from the counting term and add an overdispersion component when units differ beyond chance [17]. Population ecology makes the same split between demographic and environmental stochasticity [18, 19], as do stochastic epidemic models [20, 21]. In each case the count at which the two terms are equal has a concrete meaning: it is where a shrinkage estimator places half its weight on the direct estimate. This paper does not claim the decomposition as new. It specifies the terms for reproduction-number estimation, shows how the relevant risk changes with the target, and asks what the corresponding empirical diagnostics measure when they are computed from surveillance counts rather than from a designed experiment.

That last question matters because routine hospital surveillance does not observe the variables of a branching experiment. Admissions follow infection after a delay, aggregate heterogeneous populations and are recorded on a calendar schedule, and their count scale is an outcome of the epidemic. Reporting probabilities may change for all infections in the same period [22, 23], and separating process from observation variation in time series is a known difficulty [24, 25]. A tempting shortcut is to regress the squared relative deviation of an apparent growth statistic on inverse case count across

units and read the intercept as an irreducible error floor and the slope as counting noise. Whether such diagnostics recover those quantities has to be checked against data with a known truth.

The paper makes three contributions and one audit.

1. For a one-generation branching model with negative binomial offspring and binomial reporting, we give exact mean squared errors for the current value, the realised mean and the process mean (Section 2). The count $m_\times$ at which counting error equals between-period variance is the shrinkage half-weight point, and replacing a known denominator by the previous observed count raises the counting term.
2. When the reporting probability fluctuates between periods, single-period second moments of the counts do not separate reporting volatility from transmission variation, and the realised-mean target retains a reporting term (Section 3).
3. We define what three cross-sectional dispersion diagnostics estimate on surveillance data: the intercept as a count-independent relative dispersion, the counting slope relative to independent-Poisson and branching-Poisson references, and their ratio as a dispersion-to-counting crossover count. We apply them to 1,748 state-weeks of US influenza hospital admissions with a full sensitivity range (Section 5).

The audit applies known-truth generators to the real sampling design and shows which of these diagnostics recover their intended quantity and which cannot distinguish mechanisms (Section 4). Section 6 discusses real-time use, reporting that depends on transmission, and the choice of generation interval. Proofs and symbolic extensions are in the appendices.

2 Model and inferential targets

2.1 A branching experiment with a known observation scale

Consider one generation of transmission with $n \geq 1$ primary infections. Conditional on the reproduction number $R_t > 0$, the offspring counts of primary cases are independent negative binomial variables [26]:

$$X_{i,t} \mid R_t \sim \text{NB}(k,\, k/(k+R_t)), \qquad \text{E}(X_{i,t} \mid R_t) = R_t, \quad \text{Var}(X_{i,t} \mid R_t) = R_t + \frac{R_t^2}{k}, \tag{1}$$

where $k > 0$ controls individual heterogeneity. Each secondary infection is independently reported with probability $\rho \in (0, 1]$. Summation and binomial thinning give

$$C_t \mid n, R_t \sim \text{NB}(nk,\, k/(k+\rho R_t)), \qquad \text{E}(C_t \mid n, R_t) = \rho n R_t, \qquad \text{Var}(C_t \mid n, R_t) = \rho n R_t + \frac{\rho^2 n R_t^2}{k}. \tag{2}$$

Define $m = \rho n$ and $\hat{R}_t = C_t/m$. The scale m is the expected reported count at $R_t = 1$, not the realised count C_t.

The baseline assumptions are fixed, known ρ with independent reporting (H1); conditionally independent offspring as in Eq. (1) (H2); fixed, known k (H3); and a fixed or exogenous n that does not depend on transmission (H4). Write $\text{E}(R_t) = \overline{R} > 0$ and $\text{Var}(R_t) = v_R < \infty$. Across periods, counts are independent given the whole transmission trajectory, each with conditional law (2). These are repeated branching experiments; they do not assert that naturally evolving incidence is exogenous. Dividing by an assumed $\rho_0 \neq \rho$ gives conditional mean $(\rho/\rho_0)R_t$, a multiplicative bias that more cases do not remove.

The conditional counting variance combines offspring randomness and omission. Conditioning on the total number of secondary infections splits it into $(R_t + R_t^2/k)/n$ and $(1-\rho)R_t/(\rho n)$. The term linear in R_t is therefore not purely reporting noise; it remains under complete reporting.

2.2 Current values, realised means and process means

For the current target, $\text{E}(\hat{R}_t \mid R_t) = R_t$ and $\text{Var}(\hat{R}_t \mid R_t) = (R_t + \rho R_t^2/k)/m$, which attains the Cramér–Rao bound for estimators unbiased for every R_t under this likelihood (Appendix A.1).

Proposition 1 (One period, two targets)**.** *Under H1–H4, let $A = \overline{R} + \rho(\overline{R}^2 + v_R)/k$. Then*

$$\text{E}[(\hat{R}_t - R_t)^2] = \frac{A}{m}, \qquad V(m) := \text{E}[(\hat{R}_t - \overline{R})^2] = v_R + \frac{A}{m}. \tag{3}$$

Among estimators based on this period's count that are conditionally unbiased for R_t, $\hat{R}_t$ minimises both risks.

The proof (Appendix A.2) squares

$$\hat{R}_t - \overline{R} = (\hat{R}_t - R_t) + (R_t - \overline{R}), \tag{4}$$

Table 1 Inferential targets and exact mean squared errors. The fixed-reporting column assumes H1–H4; the volatile-reporting column assumes H2–H5 and, for averages, independent reporting probabilities independent of the transmission path. Counting errors are conditionally independent across periods. $B_T = \mathrm{Var}(\overline{R}_T)$ equals v_R/T for independent, identically distributed transmission and $v_R S(\phi, T)/T^2$ under the stated stationary autocovariance. With the previous observed count as denominator, the current-value entry under fixed reporting becomes approximately $[A + (1-\rho)(\overline{R}^2 + v_R)]/m$ (Eq. 9; Table 2).

Target	Estimator	Fixed, known reporting	Unobserved volatile reporting
Current R_t	$\hat{R}_t$	A/m	$v_H + A_H/m$
Realised mean $\overline{R}_T$	$\overline{\hat{R}}_T$	$A/(Tm)$	$(v_H + A_H/m)/T$
Process mean $\overline{R}$	$\overline{\hat{R}}_T$	$B_T + A/(Tm)$	$B_T + (v_H + A_H/m)/T$

whose two terms are uncorrelated. Between-period transmission variation is part of the signal for the current value and sampling error for the process mean, which is a mean under a stated stationary law and not automatically a seasonal average of a nonstationary epidemic.

For $v_R > 0$ define the crossover scale $m_\times = A/v_R$, the expected count at which the two terms of $V(m)$ are equal. Its meaning is the one it has in small-area estimation [14]. The best linear predictor of R_t from $\hat{R}_t$ under the stated moments is

$$\overline{R} + \gamma(m)(\hat{R}_t - \overline{R}), \qquad \gamma(m) = \frac{v_R}{v_R + A/m} = \frac{m}{m + m_\times}, \tag{5}$$

so $m_\times$ is the count at which shrinkage gives the direct estimate half its weight. Doubling m through n at fixed ρ, k and transmission law reduces process-mean risk by the fraction $1/[2(1 + m/m_\times)]$: one quarter at $m = m_\times$, and less beyond it. Changing ρ changes A as well as m. The scale $m_\times$ describes the selected estimator and target; it does not mark a point where information stops accumulating.

Corollary 1 (A fixed set of observed periods)**.** *Under H1–H4 and conditional independence of counts given $R_{1:T}$, let $\overline{R}_T$ denote the mean of $R_1, \ldots, R_T$ and $\overline{\hat{R}}_T$ the mean of $\hat{R}_1, \ldots, \hat{R}_T$. If the R_t share common marginal moments, the two mean targets have risks:*

$$\mathrm{E}[(\overline{\hat{R}}_T - \overline{R}_T)^2] = \frac{A}{Tm}, \qquad \mathrm{E}[(\overline{\hat{R}}_T - \overline{R})^2] = B_T + \frac{A}{Tm}, \qquad B_T = \mathrm{Var}(\overline{R}_T). \tag{6}$$

In Eq. (6)*, $B_T = v_R/T$ for independent, identically distributed R_t. For a stationary process with* $\mathrm{Cov}(R_t, R_{t+h}) = v_R\phi^{|h|}$*, $|\phi| < 1$, $B_T = v_R S(\phi, T)/T^2$ exactly, with*

$$S(\phi, T) = T + 2\sum\nolimits_{h=1}^{T-1}(T-h)\phi^h = \frac{T(1+\phi)}{1-\phi} - \frac{2\phi(1-\phi^T)}{(1-\phi)^2}. \tag{7}$$

The realised-mean identity needs only conditionally independent counting errors, not independent transmission levels. More cases in the same periods remove its counting error, while process-mean risk tends to B_T; only more periods reduce B_T, at a rate set by their dependence. Positive persistence reduces the gain from adjacent periods. Spatial averaging behaves analogously: for J units with transmission variance v_R and mean pairwise correlation $\bar{c}$, the averaged transmission level has variance:

$$\mathrm{Var}(J^{-1}\textstyle\sum_{j=1}^{J} R_{j,t}) = v_R[\bar{c} + (1-\bar{c})/J]. \tag{8}$$

Figure 1 plots these risks, with known-truth Monte Carlo checks, and Table 1 collects them.

2.3 An observed denominator

Surveillance ratios divide by an observed count rather than by a known scale. Let the denominator be the reported count of the primary generation, $M_t \sim \mathrm{Bin}(n, \rho)$, independent of the offspring given n, and let $\tilde{R}_t = C_t/M_t$. Because $M_t = m(1 + u_t)$ with $\mathrm{Var}(u_t) = (1-\rho)/m$, a delta expansion (Appendix A.5) gives

$$\mathrm{E}[(\tilde{R}_t - R_t)^2] \approx \frac{A + (1-\rho)(\overline{R}^2 + v_R)}{m}, \tag{9}$$

so the crossover scale moves to $m_\times + (1-\rho)(\overline{R}^2 + v_R)/v_R$. With $\overline{R} = 1.15$, $v_R = 0.04$, $k = 0.35$ and $\rho = 0.25$ ($A = 2.1232$) it rises from 53.1 to 78.6. Monte Carlo with 400,000 draws per scale (Table 2) puts the observed-denominator error at 1.59, 1.51 and 1.49 times A/m for $m = 50$, 200 and 1000, close to the delta value of 1.48. At $m = 10$ the ratio is 2.42 and the delta form understates the error, because small denominators inflate the ratio.

Table 2 Point-target error with a known versus an observed denominator

m	Known denominator A/m	MC (SE)	Observed denominator (A + (1 - ρ)($\bar{R}^2$ + v_R))/m	MC (SE)	MC observed / A/m
10	0.21232	0.21304 (0.00055)	0.31451	0.51453 (0.00482)	2.423
50	0.04246	0.04234 (0.00010)	0.06290	0.06745 (0.00019)	1.588
200	0.01062	0.01060 (0.00002)	0.01573	0.01601 (0.00004)	1.508
1000	0.00212	0.00212 (0.00000)	0.00315	0.00316 (0.00001)	1.489

Parameters $\bar{R}$ = 1.15, v_R = 0.04, k = 0.35, ρ = 0.25 (A = 2.1232); 400,000 Monte Carlo draws per m. Crossover scale $m_\times$ = A/v_R = 53.08 with a known denominator and (A + (1 - ρ)($\bar{R}^2$ + v_R))/v_R = 78.63 when the denominator is the observed previous count (delta approximation). At m = 10 the delta approximation understates the Monte Carlo error of the observed-denominator estimator.

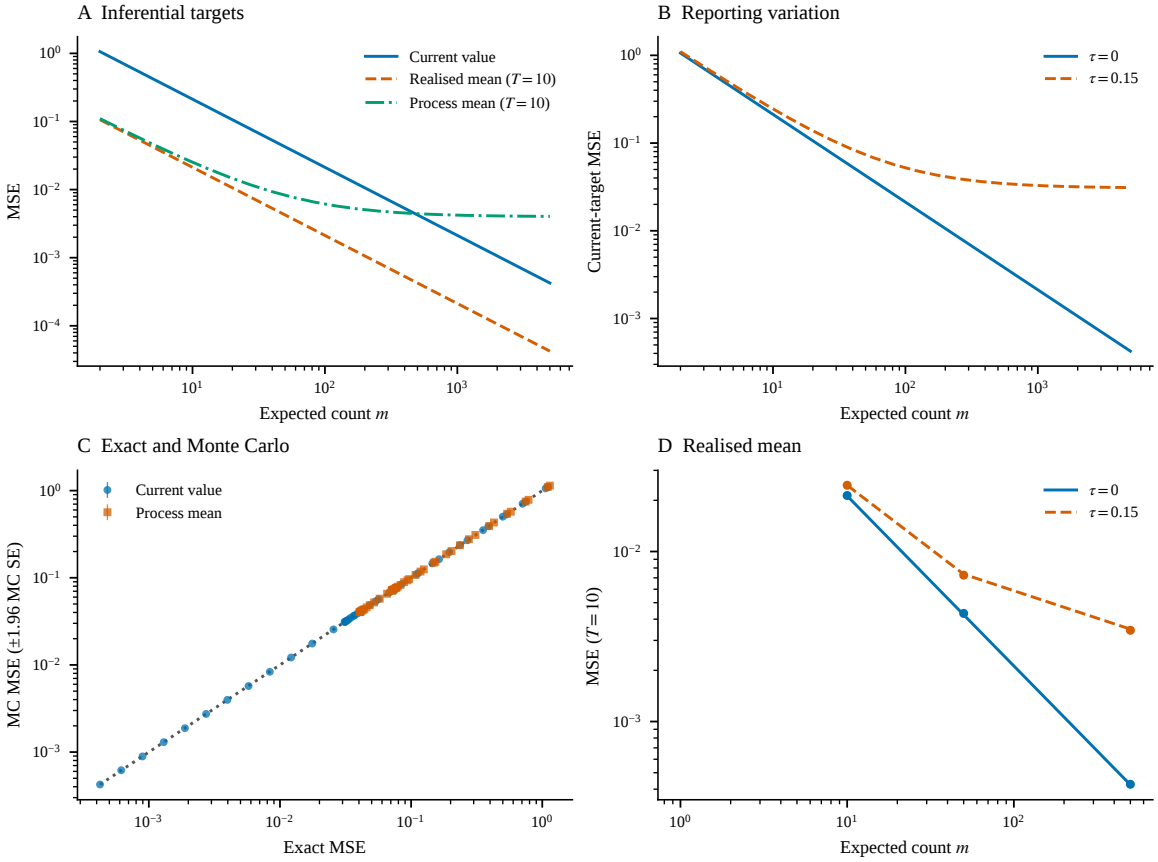


Figure 1 Model risks for the three targets. (A) Mean squared error against expected count m for the current value, the realised mean and the process mean over $T = 10$ independent periods. (B) Current-value risk with fixed reporting ($\tau = 0$) and with period-level reporting variation ($\tau = 0.15$). (C) Exact risks against known-truth Monte Carlo estimates with 1.96 Monte Carlo standard errors. (D) Realised-mean risk over $T = 10$ periods at three scales. Parameters: $\overline{R} = 1.15$, $v_R = 0.04$, $k = 0.35$, $\rho = 0.25$. Counts are simulated under the fixed-exposure model, so the curves describe that model and not the hospital data.

3 Reporting volatility and identification

3.1 Period-level reporting variation

Replace H1 by a common period-specific reporting probability $p_t \in [0,1]$, independent of R_t and of n, with known mean $\rho > 0$ and $\mathrm{Var}(p_t) = \rho^2\tau^2$ (H5). Reporting remains independent across infections given p_t, so $\tau^2 \le (1-\rho)/\rho$. The estimator still divides by $m = \rho n$ and does not observe p_t.

Proposition 2 (Reporting volatility and second-moment nonidentification)**.** *Under H2–H5, put* $Q = \overline{R}^2 + v_R$. *Then*

$$\mathrm{E}(\hat{R}_t) = \overline{R}, \qquad \mathrm{Var}(\hat{R}_t) = v_R + v_H + \frac{A_H}{m}, \qquad v_H = \tau^2 Q, \qquad A_H = \overline{R} + \frac{\rho(1+\tau^2)Q}{k}. \tag{10}$$

For known m, ρ *and* k, *the single-period mean and variance in Eq.* (10) *depend on* v_R *and* τ^2 *only through* $U = (1+\tau^2)(\overline{R}^2 + v_R)$.

Given R_t and p_t the estimator has mean $(p_t/\rho)R_t$, so a shared reporting shock is not averaged away by more infections in the same period. The statement is deliberately narrow. It concerns the first two moments of one period's count within the specified family. It does not exclude identification from full count distributions, from temporal structure under further assumptions, or from external measurements of reporting; it also assumes that reporting is independent of transmission, an assumption Section 6 relaxes.

Corollary 2 (Current and realised-mean risks with volatile reporting)**.** *Under H2–H5, the current-value risk is:*

$$\mathrm{E}[(\hat{R}_t - R_t)^2] = v_H + \frac{A_H}{m}. \tag{11}$$

If, in addition, the p_t *are independent and identically distributed, independent of the transmission trajectory, and counts are independent given both trajectories, then*

$$\mathrm{E}[(\overline{\hat{R}}_T - \overline{R}_T)^2] = \frac{v_H + A_H/m}{T}, \qquad \mathrm{E}[(\overline{\hat{R}}_T - \overline{R})^2] = B_T + \frac{v_H + A_H/m}{T}. \tag{12}$$

At fixed T, unobserved reporting volatility leaves the floor v_H/T on the realised-mean target as m grows, so a target free of transmission variation is not free of reporting error. The crossover scale with volatile reporting, $m_{\times,H} = A_H/(v_R+v_H)$, is never larger than $m_\times$ (Eq. (A.5)); the lower crossover accompanies a larger risk, not better surveillance. Proofs are in Appendix A.4.

3.2 From the branching experiment to surveillance counts

If the scale m is random and associated with transmission, conditional unbiasedness given (R_t, m) still yields

$$\mathrm{E}[(\hat{R}_t - \overline{R})^2 \mid m] = \frac{\mathrm{E}(R_t \mid m) + \rho\,\mathrm{E}(R_t^2 \mid m)/k}{m} + \mathrm{Var}(R_t \mid m) + [\mathrm{E}(R_t \mid m) - \overline{R}]^2, \tag{13}$$

and the conditional moments in Eq. (13) can change with scale. A constant intercept plus a constant coefficient on inverse count therefore does not follow when the count scale is itself an epidemic outcome, as it is in surveillance. Delays, aggregation, ratio denominators and changing conditional means further separate an apparent statistic from the experiment above. The rest of the paper accordingly treats empirical dispersion diagnostics as descriptive quantities and asks, with a known-truth audit, what they recover.

4 What cross-sectional dispersion diagnostics measure

4.1 Data and apparent statistics

We used the National Healthcare Safety Network (NHSN) weekly hospital respiratory-data snapshot retrieved on 20 October 2026 and restricted influenza admissions to the 50 US states and the District of Columbia [27, 28]. Hospital participation varied during the series, including a voluntary-reporting interval from May to October 2024 before reporting became mandatory again [29]. The analysis describes this surveillance system and does not estimate an effect of the policy change.

The windows were the early-growth phases of four seasons: 1 October–26 November 2022, 7 October–23 December 2023, 9 November–28 December 2024 and 4 October–27 December 2025. Each is the contiguous national segment with weekly growth of at least 1.05 before the first turning point after the season's admissible start; the 2024 start follows the return to mandatory reporting. The window rule was inherited from the original analysis pipeline and was not tuned here;

inference conditions on the selected windows. Missing or suppressed counts were kept missing rather than set to zero. A row requires four observed counts on consecutive weekly dates, from two weeks before its denominator week through the following week.

With C_t the admissions in week t, we used two apparent statistics. The weighted growth ratio is

$$G_t = \frac{C_{t+1}}{D_t}, \qquad D_t = 0.65C_t + 0.25C_{t-1} + 0.10C_{t-2}, \tag{14}$$

and the growth-to-reproduction transform is

$$R_{A,t} = [\max\{0,\, 1 + (\theta/7)\log\{(C_{t+1} + 0.5)/(C_t + 0.5)\}\}]^{\alpha}, \qquad \alpha = (g_\mu/g_\sigma)^2, \quad \theta = g_\sigma^2/g_\mu, \tag{15}$$

which follows the renewal relation between growth rate and reproduction number for a gamma-distributed interval [30]. Neither statistic is a true reproduction number or a posterior estimate from a renewal model [3, 10]. Rows require $D_t \geq 5$ and, for R_A, $C_t \geq 5$, which leaves 1,748 state-weeks for R_A and 1,756 for G.

The primary interval has mean $g_\mu = 3.6$ days and standard deviation $g_\sigma = 1.6$ days, the household serial-interval estimate of Cowling et al. [31]. It was kept as the primary choice because it is directly estimated from influenza household transmission and is widely used in renewal-based influenza work, even though a serial interval is not a generation interval [32]. Two sensitivities use a mean of 3.2 days, either with the same coefficient of variation or with standard deviation 2.1 days; the latter is the intrinsic generation-time estimate from US household data [33]. Section 5.3 shows how much the choice matters.

4.2 The diagnostic and its three summaries

For an apparent statistic X with pooled mean $\bar{X}$, define the relative squared deviation and its affine working model

$$L_{st} = \frac{(X_{st} - \bar{X})^2}{\bar{X}^2}, \qquad \mathrm{E}(L_{st} \mid C_{st}) = a + \frac{b}{C_{st}}, \tag{16}$$

with D_t of Eq. (14) in place of C_t for G. This is the empirical counterpart of $V(m)/\overline{R}^2$ in Eq. (3), but with a realised count in place of the expected scale and an apparent statistic in place of $\hat{R}_t$. Three summaries of the fit of Eq. (16) follow.

Count-independent relative dispersion, a. The intercept measures the sum of all relative variation in the diagnostic statistic that does not scale with count. When reporting probabilities do not vary from week to week ($\tau = 0$), a estimates the relative variance of transmission, $\tilde{v}_R = v_R/\overline{R}^2$. When reporting shocks vary between weeks, a estimates $\tilde{v}_R + \tilde{v}_H$, where $\tilde{v}_H$ is the relative variance that the shocks induce in the ratio statistic; a reporting level that is constant over consecutive weeks cancels in the ratio (Appendix A.6). By Proposition 2, single-period cross-sectional moments cannot separate the two components. The intercept is therefore not an estimate of an error floor: with $\tau = 0$ the current-value error floor is zero, and with $\tau > 0$ the floor $\tilde{v}_H$ does not exceed a, which is only an upper bound for it. Below we call a the count-independent relative dispersion, or the relative dispersion for short.

Counting slope, b, against two references. A delta expansion of Eq. (15) under independent Poisson counts in consecutive weeks gives the independent-Poisson reference $b_{\mathrm{P}} = \overline{\kappa(g)^2[1 + (C_t + 0.5)/(C_{t+1} + 0.5)]}$, a weighted mean over rows, with $\kappa(g) = (g_\mu/7)/[1 + \theta\log(g)/7]$ and $g = (C_{t+1} + 0.5)/(C_t + 0.5)$. For the primary sample $b_{\mathrm{P}} = 0.4455$. The term 1 is the Poisson variance of the denominator count. In a branching chain the denominator week contains the parents of the numerator week, so conditioning on the realised parents removes the transmission part of that variance and leaves only the binomial reporting part, $(1 - \rho)/C_t$. With Poisson offspring and reporting probability ρ this gives the branching-Poisson reference

$$b_{\mathrm{P,br}}(\rho) = \overline{\kappa(g)^2[(1 - \rho) + (C_t + 0.5)/(C_{t+1} + 0.5)]}, \qquad r(\rho) = \frac{b_{\mathrm{P,br}}(\rho)}{b_{\mathrm{P}}} \approx \frac{1 + (1 - \rho)g}{1 + g}, \tag{17}$$

where the row-level ratio $r(\rho)$ falls to $1/(1 + g) \approx 0.42$ at $\rho = 1$ for the geometric-mean growth $g_0 = 1.407$. Averaged over the observed rows, $r(\rho)$ is 0.944, 0.861, 0.722 and 0.443 for $\rho = 0.1$, 0.25, 0.5 and 1 (Table S2). The two references correspond to the two estimators of Section 2. The observed-denominator model of Section 2.3, whose denominator noise $(1 - \rho)/m$ is the reporting part only, has the structure of the branching-Poisson reference, and the known-scale estimator of Section 2.1 is its limit without denominator noise. The independent-Poisson reference is the limit $\rho \to 0$, in which the denominator is an independent Poisson count. The ratio b/b_{P} therefore asks whether counting variation exceeds independent-Poisson noise, and $b/b_{\mathrm{P,br}}(\rho)$ asks the same question for a branching chain, at a reporting probability that the data do not identify.

Dispersion-to-counting crossover count, b/a. The count at which the fitted counting term equals the relative dispersion is the empirical analogue of $m_\times$. It is defined on the apparent statistic's count scale, in admissions per week, and differs from $m_\times$, which refers to the expected count of a branching experiment.

The affine mean was fitted by iterated feasible generalised least squares with weights inversely proportional to the squared fitted mean, which minimises a gamma quasi-deviance; this is an estimating criterion, not a distributional claim.

Intervals are percentiles of state-by-week pigeonhole bootstrap draws in which each row receives the product of independently drawn state and week multiplicities [34]; the pooled mean and the regression are re-estimated in every draw. The primary bootstrap has 2,000 draws, and every pooled row in every table uses its weights. The within-state and two-way slopes, the season models, the generation-interval sensitivity and the sensitivity specifications also use 2,000 draws. The two- and three-week block versions of the fixed-effects decomposition, which check sensitivity to persistent national shocks, use 1,000 draws each, as do the thinning analysis of the data and the held-out validation. Draws whose iterated fit does not converge are dropped, and their number is reported with each table.

Binomial thinning. Each unique state-week count is thinned once to $\mathrm{Bin}(C, q)$ for $q \in \{0.8, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1\}$ and the thinned value is reused in every ratio that contains it. Recomputing the statistic and the fit with the original C_t as regressor gives

$$b(q) - b(1) \simeq c\,(1/q - 1), \tag{18}$$

where under the delta approximation $c = b_{\mathrm{P}}$ whatever the original counting variation, because thinning adds known binomial variance. The ratio c/b_{P} estimated from Eq. (18) is therefore a check of that approximation, not of the data-generating process.

4.3 Known-truth audit on the real design

To learn what a, b, b/a and c recover, we simulated ten generators on the exact R_A design: the 1,748 rows, 1,969 unique state-weeks and 210 state runs of the application. Each run is simulated as a whole with shared counts, so that one week's count is the next week's denominator. The expected count is carried forward by the true growth factors, with the geometric mean observed growth $g_0 = 1.407$ used for transitions not scored. The simulated C_t serves as the count scale, the filter $C_t \geq 5$ is re-applied, and the target R_i of each row is the transform (15) of its true growth factor.

Simulated counts were aligned with the observed ones in two steps. Each run is level-anchored to the geometric mean of its observed counts, and a global scale factor per generator, between 0.797 and 0.844, sets the Monte Carlo mean of the median retained count to 45, the observed median. The lower tail then matches the data (10th percentiles 10–11 against 10), but the upper tail does not: the generators' 90th percentiles are 263–324 against 389 observed (Table 3). Reweighting synthetic panels to match the empirical upper-tail representation shifts intercept recovery across S1, S4, S6 and S7a by at most 0.004–0.043 (<4%), confirming that tail truncation does not alter the recovery assessment. In every generator with transmission heterogeneity, its amplitude was then set so that the Monte Carlo mean of the fitted intercept equals the observed a. The generators are defined as follows.

- S0: constant growth g_0 with independent Poisson counts.
- S1: independent lognormal growth factors ($\sigma = 0.247$) with Poisson counts.
- S2: constant growth with negative binomial offspring ($k = 0.3$) and binomial reporting at $\rho = 0.25$.
- S2P: S2 with Poisson offspring, a pure branching chain with reporting.
- S3: S1 combined with the S2 counting model.
- S4: growth heterogeneity split equally between a state and a national-week component, with Poisson counts.
- S5: S4 combined with the S2 counting model.
- S6: S3 with a state-week reporting shock, $\rho_{st} = 0.25 z_{st}$ with $\log z_{st} \sim N(-s^2/2, s^2)$ and $s = 0.122$; transmission and the shock each contribute about half of the observed a.
- S7a and S7b: S3 with scale-dependent heterogeneity, $\sigma^2(\mu) = \sigma_0^2(\mu/45)^{-\gamma}$ for expected count μ, with $\gamma = 0.25$ and 0.5.

Each generator was run on 200 panels, so the quantile of an observed value among simulated values has a resolution of 0.005. Thinning used 100 panels with 20 draws per level. Every panel passes through the same filters and fitting functions as the data. The audit covers R_A; the G statistic was not audited.

4.4 Audit results

Without reporting shocks the intercept tracks the relative variance of transmission. When transmission is constant (S0, S2, S2P) the fitted intercept is essentially zero, so b/a is undefined. With independent heterogeneity, the ratio of a to the true relative variance of R_i on the retained rows is 1.01 (Monte Carlo range 0.93–1.09) under S1 and 1.02 (0.91–1.13) under S3; with structured heterogeneity it is 1.12 (1.04–1.19) under S4 and 1.15 (1.06–1.27) under S5 (Table 3; Fig. 2C). Fitting the same affine model to the true estimation error, $((X - R_i)/\bar{X})^2$, gives an intercept of about 3×10^{-5} in each of these generators, so without reporting shocks the current-value error floor is essentially zero. Week centring removes 61.8% of a under S4 and 58.6% under S5, where half of the heterogeneity is national, and essentially none under S1 and S3.

With reporting shocks the intercept is the sum of both components. Under S6, a is 1.01 (0.87–1.14) times $\tilde{v}_R + \tilde{v}_H$ and 2.02 (1.71–2.29) times $\tilde{v}_R$ alone. The true-error intercept is now 1.02 (0.83–1.19) times $\tilde{v}_H$, so the current-value error

floor equals the shock component and is about half of a. The shock terms of consecutive rows have correlation -0.50 (-0.53 to -0.46), because a shock in one week enters the numerator of one row and the denominator of the next. Scale-dependent heterogeneity breaks the intercept in the other direction: under S7a and S7b the intercept is 0.68 (0.60–0.78) and 0.43 (0.37–0.50) times the true relative variance, because variation that shrinks with count is absorbed by the slope.

The counting slope does not identify the mechanism. The ratio b/b_{P} is 0.93 (0.85–1.01) under S0, 0.80 (0.73–0.86) under S2P and 1.23 (1.13–1.32) under S2 (Table 4; Fig. 2A). The step from S0 to S2P is the denominator-noise structure: in a branching chain only the reporting part of the denominator noise remains, as the branching-Poisson reference predicts ($r(0.25) = 0.861$; simulated 0.802, Table S2). Normalizing by the S0 recovery factor (0.930, reflecting the common finite-sample downward bias of gamma FGLS under this design) brings the simulated reference into exact alignment with the analytic prediction ($0.802/0.930 = 0.862$). The step from S2P to S2 is offspring overdispersion, $k = 0.3$ in place of Poisson offspring. Heterogeneity in transmission lowers the ratio: 0.85 (0.69–1.02) under S1, 1.14 (0.98–1.31) under S3, 0.73 (0.60–0.87) under S4 and 1.03 (0.87–1.20) under S5, with 1.14 (0.97–1.34) under S6. Scale-dependent heterogeneity raises it, to 1.56 (1.34–1.79) under S7a and 2.25 (1.98–2.58) under S7b. A slope near or above b_{P} is therefore compatible with several combinations of overdispersion, reporting and heterogeneity. Structured heterogeneity also lowers the within-state slope below the pooled slope although counting noise is the same in every state: under S4 the ratio of pooled to within-state slope is 1.14 (0.99–1.37), and in the other generators it lies between 1.01 and 1.10. The observed ratio, 1.48 (0.99–3.66), is larger than any of these but has a lower limit near 1.

The crossover count does not discriminate between mechanisms either. Among generators with heterogeneity, b/a is 25.9 (19.4–32.9) under S1, 36.3 (27.4–44.4) under S3, 22.2 (17.0–27.7) under S4, 32.6 (24.7–40.0) under S5 and 36.4 (28.2–47.6) under S6, against 51.4 (39.5–65.9) and 77.5 (58.4–99.7) under S7a and S7b. The observed value, 37.4, lies within the S3, S5 and S6 envelopes, outside those of S1, S4, S7a and S7b, and its bootstrap interval, 18.4–72.4, overlaps all seven.

Only scale-dependent heterogeneity reproduces the curve shape. The in-sample difference in quasi-deviance score between affine and power-law fits is close to zero (-0.0004 to -0.0003) under S0, S2 and S2P, whose intercept is zero. The generators with constant heterogeneity give differences from 0.012 (S1) to 0.024 (S5), in favour of the affine form, whereas the observed difference is -0.0014, at or below the 1% quantile of each of them (Fig. 2D). Under S7a the mean contrast is -0.0003 and the observed value lies at its 43% quantile; under S7b every simulated contrast lies below the observed one. S7a thus matches both the observed intercept, by construction, and the observed curve shape, and its slope ratio interval (1.34–1.79) overlaps the observed one. Combining moderate scale-dependent heterogeneity ($\gamma = 0.25$) with period-level reporting volatility ($s = 0.133$, as in S6) simultaneously accommodates all three summaries: the fitted intercept is 0.0148, the crossover count is 41.7 (30.6–52.1, covering the observed 37.4), the slope ratio is 1.298 (covering 1.257), and the observed score contrast falls at the 8.0% quantile of the simulated distribution. One generator that fits does not establish the mechanism, and the audit bounds what the summaries recover rather than identifying a best generator.

Table 3 Known-truth audit on the real R_A design: retained counts, intercept, true error and recovery by generator

Generator	n retained	D median [P10, P90]	True rel. variance (retained rows)	Intercept a	a / truth	True-error intercept	Week-centred a reduction
Observed (R_A, GI 3.6)	1748	45 [10, 389]	unknown	0.0150 [0.0101, 0.0206]	–	–	0.35
S0 Poisson, no heterogeneity	1671	45 [10, 281]	0	0.0000 [-0.0001, 0.0002]	truth 0	0.00003 [-0.00010, 0.00019]	-30.74 [-34.19, 12.89]
S1 iid heterogeneity, Poisson counts	1678	45 [11, 263]	0.0150	0.0151 [0.0139, 0.0165]	1.01 [0.93, 1.09]	0.00003 [-0.00010, 0.00021]	-0.01 [-0.04, 0.01]
S2 NB offspring + binomial reporting, no heterogeneity	1582	45 [10, 324]	0	0.0001 [-0.0001, 0.0003]	truth 0	0.00002 [-0.00013, 0.00018]	0.44 [-20.56, 35.68]
S3 iid heterogeneity + NB/reporting	1592	45 [10, 298]	0.0147	0.0150 [0.0133, 0.0168]	1.02 [0.91, 1.13]	0.00003 [-0.00011, 0.00020]	-0.02 [-0.05, 0.01]
S4 state + week heterogeneity, Poisson counts	1658	45 [11, 315]	0.0135	0.0150 [0.0140, 0.0161]	1.12 [1.04, 1.19]	0.00002 [-0.00006, 0.00015]	0.62 [0.58, 0.66]
S5 state + week heterogeneity + NB/reporting	1555	45 [10, 324]	0.0130	0.0149 [0.0136, 0.0166]	1.15 [1.06, 1.27]	0.00003 [-0.00007, 0.00017]	0.59 [0.53, 0.64]
S2P Poisson offspring + binomial reporting, no heterogeneity	1655	45 [10, 293]	0	0.0000 [-0.0001, 0.0001]	truth 0	0.00002 [-0.00008, 0.00015]	-2.64 [-65.20, 22.66]
S6 S3 + state-week reporting shock (half of a)	1587	45 [10, 307]	0.0147	0.0148 [0.0127, 0.0168]	1.01 [0.87, 1.14]	0.00746 [0.00590, 0.00877]	-0.01 [-0.05, 0.02]
S7a S3 with scale-dependent heterogeneity, gamma 0.25	1603	45 [10, 296]	0.0217	0.0148 [0.0130, 0.0169]	0.68 [0.60, 0.78]	0.00003 [-0.00011, 0.00020]	-0.03 [-0.08, 0.02]

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(continued)

Generator	n retained	D median [P10, P90]	True rel. variance (retained rows)	Intercept a	a / truth	True-error intercept	Week-centred a reduction
S7b S3 with scale-dependent heterogeneity, gamma 0.5	1614	44 [11, 286]	0.0346	0.0150 [0.0130, 0.0173]	0.43 [0.37, 0.50]	0.00002 [-0.00013, 0.00021]	0.00 [-0.08, 0.06]
S8 Combo S6 shock + S7a scale-dep ($\gamma = 0.25$)	1598	45 [10, 301]	0.0215	0.0148 [0.0128, 0.0169]	0.69 [0.61, 0.79]	0.00742 [0.00588, 0.00874]	-0.02 [-0.07, 0.02]

200 synthetic panels per generator on the real 1748-row R_A design; analyst scale D = simulated C_t and the filter $C_t \geq 5$ applied to each panel; n retained and the D quantiles are Monte Carlo means over panels. Count alignment: each run of consecutive weeks is level-anchored to the observed counts of that run, and a global scale per generator (0.797-0.844) sets the mean retained median D to 45 (observed 45). The upper tail is not matched: the generator P90 of D is 263-324 against the observed 389. Intervals are Monte Carlo 2.5-97.5% ranges across panels; a / truth is the ratio of panel means with the range of panel ratios. Truth is the relative variance of the target R_i on retained rows; for S6 it also includes the relative variance of the reporting-shock term. True-error intercept: affine fit of $((x - R_i)/\bar{x})^2$. Week-centred reduction = 1 - a_{week}/a. Observed a: primary bootstrap (2,000 draws). S6 (shock log-SD 0.122): a / v_R = 2.02 [1.71, 2.29]; true-error intercept / reporting-shock variance = 1.02 [0.83, 1.19]; lag-1 correlation of the shock term between consecutive rows -0.50 [-0.53, -0.46].

Table 4 Known-truth audit: b/a, slope references, thinning and the in-sample contrast by generator

Generator	b/a	b/b_{P}	$b_{\text{w}}/b_{\text{P}}$ (state FE)	Pooled / within slope	Thinning c/b_{P}	In-sample contrast	Quantile of observed contrast
Observed	37.4 [18.4, 72.4]	1.257 [0.812, 1.734]	0.852 [0.237, 1.475]	1.48 [0.99, 3.66]	0.909 [0.810, 0.999]	-0.0014	–
S0 Poisson, no heterogeneity	—	0.930 [0.853, 1.005]	0.896 [0.794, 1.021]	1.04 [0.95, 1.13]	0.970 [0.946, 1.000]	-0.0004 [-0.0028, 0.0018]	0.200
S1 iid heterogeneity, Poisson counts	25.9 [19.4, 32.9]	0.855 [0.686, 1.020]	0.819 [0.617, 1.026]	1.05 [0.94, 1.21]	0.950 [0.912, 0.986]	0.0119 [0.0035, 0.0213]	0.000
S2 NB offspring + binomial reporting, no heterogeneity	—	1.225 [1.128, 1.321]	1.206 [1.064, 1.342]	1.02 [0.95, 1.10]	0.929 [0.898, 0.970]	-0.0003 [-0.0033, 0.0031]	0.190
S3 iid heterogeneity + NB/reporting	36.3 [27.4, 44.4]	1.143 [0.983, 1.308]	1.123 [0.899, 1.333]	1.02 [0.93, 1.14]	0.879 [0.828, 0.927]	0.0164 [0.0044, 0.0281]	0.010
S4 state + week heterogeneity, Poisson counts	22.2 [17.0, 27.7]	0.734 [0.602, 0.871]	0.649 [0.486, 0.819]	1.14 [0.99, 1.37]	0.997 [0.964, 1.035]	0.0123 [0.0032, 0.0229]	0.000
S5 state + week heterogeneity + NB/reporting	32.6 [24.7, 40.0]	1.025 [0.866, 1.201]	0.953 [0.743, 1.142]	1.08 [0.94, 1.25]	0.925 [0.875, 0.981]	0.0239 [0.0128, 0.0363]	0.000
S2P Poisson offspring + binomial reporting, no heterogeneity	—	0.797 [0.732, 0.863]	0.774 [0.684, 0.865]	1.03 [0.96, 1.13]	0.974 [0.953, 1.004]	-0.0004 [-0.0037, 0.0022]	0.205
S6 S3 + state-week reporting shock (half of a)	36.4 [28.2, 47.6]	1.144 [0.970, 1.342]	1.141 [0.935, 1.354]	1.01 [0.90, 1.12]	0.869 [0.814, 0.919]	0.0176 [0.0048, 0.0337]	0.000
S7a S3 with scale-dependent heterogeneity, gamma 0.25	51.4 [39.5, 65.9]	1.564 [1.337, 1.787]	1.484 [1.249, 1.759]	1.06 [0.95, 1.18]	0.914 [0.861, 0.986]	-0.0003 [-0.0112, 0.0106]	0.430
S7b S3 with scale-dependent heterogeneity, gamma 0.5	77.5 [58.4, 99.7]	2.251 [1.975, 2.582]	2.058 [1.656, 2.493]	1.10 [0.97, 1.21]	0.922 [0.862, 0.980]	-0.0142 [-0.0225, -0.0046]	1.000
S8 Combo S6 shock + S7a scale-dep ($\gamma = 0.25$)	41.7 [30.6, 52.1]	1.298 [1.085, 1.542]	1.231 [0.994, 1.512]	1.05 [0.94, 1.18]	0.892 [0.835, 0.951]	0.0103 [-0.0031, 0.0245]	0.080

b_{P} is the independent-Poisson reference computed on each panel. S2, S3, S5, S6 and S7 use negative-binomial offspring (k = 0.3) with binomial reporting probability 0.25; S2P uses Poisson offspring with the same reporting. b/a undefined when a ≈ 0 (S0, S2, S2P: true relative variance 0), shown as —. Pooled / within slope = (b/b_{P})/($b_{\text{w}}/b_{\text{P}}$) with state fixed effects. Observed row: primary bootstrap (2,000 draws); its pooled / within slope is the slope ratio of the fixed-effects decomposition. Thinning uses 100 panels × 20 shared-count binomial draws per q and the least-squares rate through the origin. Contrast = in-sample affine minus power mean score; its last column is the share of synthetic panels with a contrast at or below the observed value.

4.5 Thinning as a check of the delta approximation

The exact expected thinned loss can be computed by summing over binomial supports, including the continuity correction and the positive part in Eq. (15). The exact increment rate $[b(q) - b(1)]/(1/q - 1)$ is 0.467 at $q = 0.8$, rises to 0.505 at $q = 0.3$ and then falls to 0.480 at $q = 0.2$ and 0.384 at $q = 0.1$ (Table S3; Fig. 3). The least-squares rate through the origin is 0.408, or 0.917 b_{P}, matching the Monte Carlo estimate on the data, 0.405 (0.366–0.441), or 0.909 (0.810–0.999) b_{P} (Table S4). The level $q = 0.1$ carries 76% of the leverage; without it the exact rate is 0.488, or 1.096 b_{P}. The shortfall has a simple source. At $C = 5$ and $q = 0.1$ the exact added variance of $\log(\text{Bin} + 0.5)$ is only 0.21 of its delta value $(1 - q)/(qC)$ (Table S5), so the delta approximation fails at the most thinned level; the positive-part truncation never binds. A thinning rate below b_{P} is therefore mainly the small-count delta bias of the log transform. Thinning also cannot separate offspring overdispersion from under-reporting, because it adds binomial reporting noise whatever the source of the existing counting variation. Its baseline does shift with the generator: the Monte Carlo range of c/b_{P} is 0.95–1.00 under S0, overlaps it under S4 (0.96–1.04) and S5 (0.87–0.98), and lies entirely below it under S3 (0.83–0.93) and S6 (0.81–0.92) (Fig. 2B). The observed rate lies outside the ranges of the four generators with Poisson offspring (S0, S1, S2P and S4) and inside those of the six with negative binomial offspring. Its bootstrap interval, however, overlaps every range, so the thinning rate alone does not single out a mechanism.

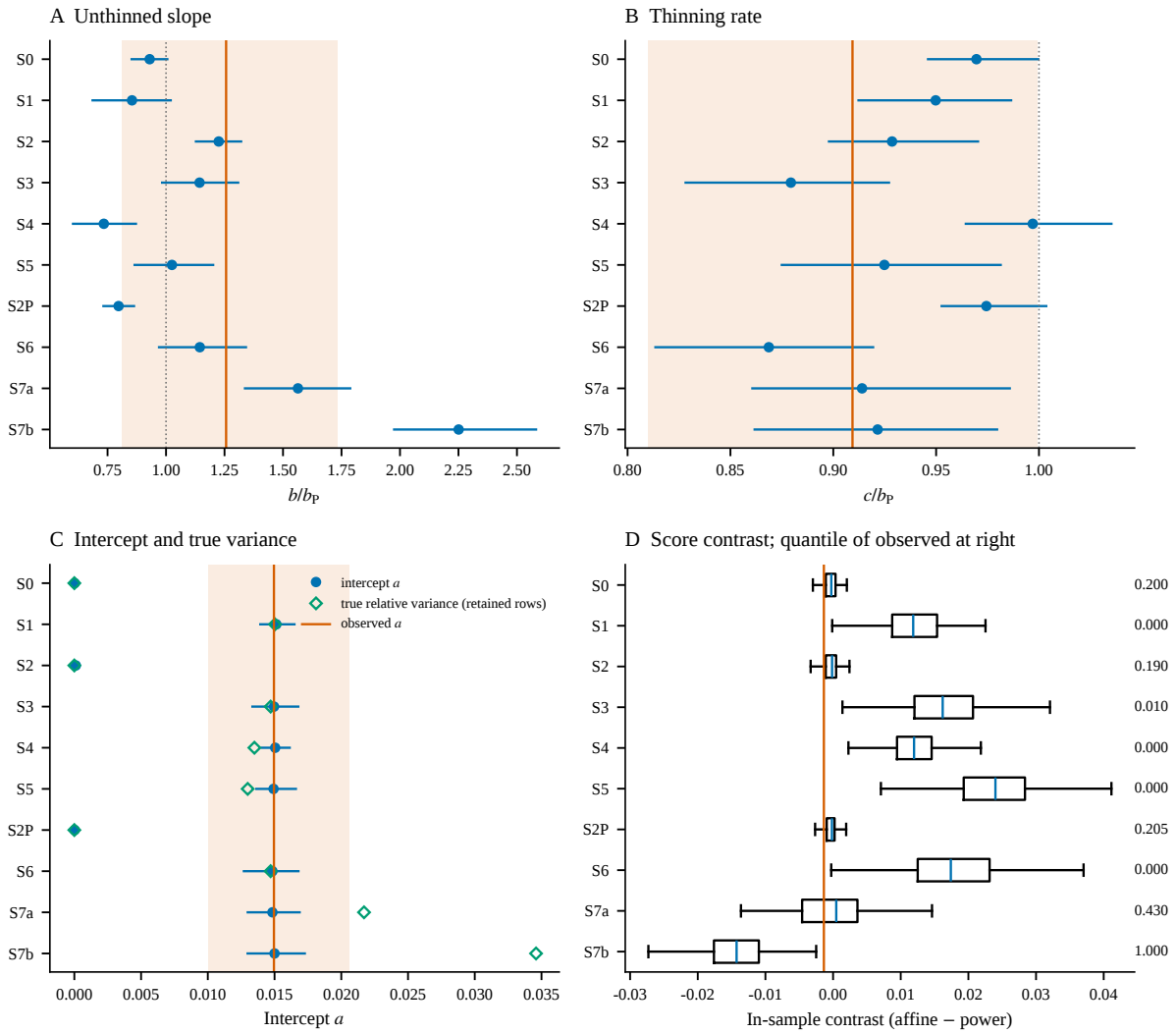


Figure 2 Known-truth audit of the dispersion diagnostics on the real R_A design, with simulated counts aligned to the observed median. (A) Counting slope relative to the independent-Poisson reference, b/b_P, by generator. (B) Thinning rate relative to the same reference, c/b_P. (C) Fitted intercept a and the true relative variance on the retained rows (for S6 including the reporting-shock component). (D) In-sample score contrast between the affine and power-law fits, with the quantile of the observed value among synthetic panels at the right; boxes span the interquartile range and whiskers 1.5 times it. In (A)–(C) points are means over 200 panels per generator (100 panels for thinning) and bars are 2.5–97.5% Monte Carlo ranges. The vertical line and shaded band mark the observed value and its bootstrap 95% confidence interval; dotted lines mark 1. S0: Poisson; S1: independent heterogeneity; S2: negative binomial offspring with reporting; S2P: Poisson offspring with reporting; S3: S1 with S2; S4: state and national-week heterogeneity; S5: S4 with S2; S6: S3 with week-to-week reporting shocks; S7a, S7b: S3 with heterogeneity that shrinks with count.

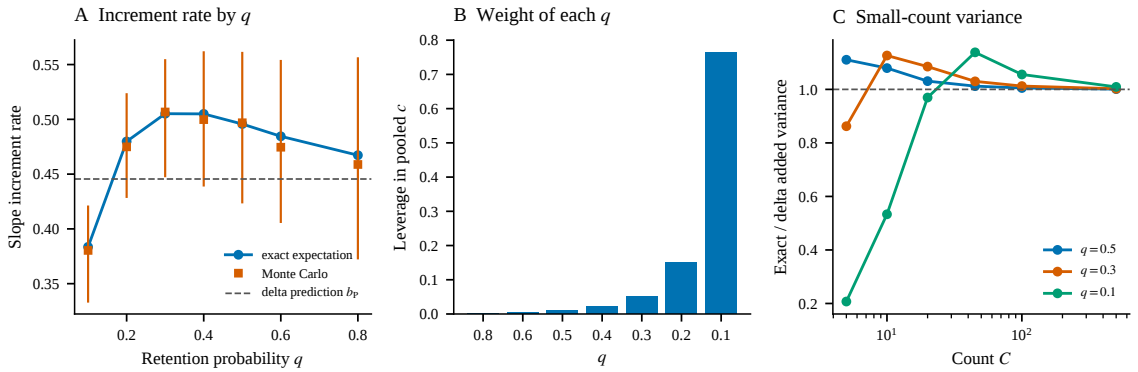


Figure 3 Exact expected thinning response of the R_A statistic on the observed design. (A) Increment rate $[b(q) - b(1)]/(1/q - 1)$ by retention probability: exact expectation from summation over binomial supports, and the Monte Carlo estimate on the data with its 95% bootstrap interval; the dashed line is the delta-method prediction $b_P = 0.4455$. (B) Share of the least-squares rate through the origin carried by each level. (C) Exact added variance of the log thinned count relative to its delta value, by original count, for three retention probabilities. The fall in (A) at $q \leq 0.2$ follows from the small-count failure shown in (C).

5 Application to US influenza hospital admissions

5.1 Count-independent relative dispersion

For R_A the pooled affine fit gave $a = 0.0150$ (95% confidence interval 0.0101–0.0206) and $b = 0.560$ (0.362–0.769) (primary row of Table 5). The count-independent relative dispersion a corresponds to a relative SD of about 12% around the pooled mean of R_A, an upper bound for the relative SD of transmission under the audited mechanisms when reporting varies between weeks (Section 4.2). The renewal growth statistic G is reported in Appendix Table S1 as an un-audited descriptive comparison on the weighted denominator D_t.

Week centring, which replaces the pooled mean by the national mean of each calendar week, reduced a by 34.7% (6.0–51.9) for R_A and by 33.3% (6.8–51.6) for G (Table S6; Fig. S1). Leaving the own state out of the weekly centre gave nearly the same values. Season means and season-specific linear trends left a almost unchanged. In the audit, week centring removed almost nothing when transmission heterogeneity was independent across states (-2.8% to 0.4% in S1, S3, S6, S7a and S7b) and removed 61.8% (S4) and 58.6% (S5) when half of it was national. The observed reduction lies between these values, which is consistent with a partly synchronous national component of intermediate size; no generator was built with that share. Adjusting for hospital reporting participation changed a by 0.00029 (-0.00032 to 0.00085) (Table S7). Off-season summer weeks, outside the analysis windows, are described in Table S12: reporting was voluntary only in summer 2024, when median participation was 45.6% against 92.3–94.2% in the other two summers and the interquartile range of $\log G$ was 0.67 against 0.49; this contrast is descriptive.

5.2 Counting slope

The pooled counting slope was 1.26 (0.81–1.73) times the independent-Poisson reference: 26% above it, with an interval that includes 1. Against the branching-Poisson reference of Eq. (17) the ratio was 1.46 (0.94–2.02) at $\rho = 0.25$ and 2.84 (1.83–3.94) at $\rho = 1$ against the uncorrected simulated reference, corresponding to 1.36 and 2.64 when adjusted for the S0 finite-sample bias (Table S2). The excess over a branching-Poisson reference therefore depends on the reporting probability, which the slope itself does not identify.

Within-unit estimates were lower and less precise (Table S8; Fig. 5B). A state fixed-effects fit gave 0.85 (0.24–1.47) b_{P} by iterated weighting and 1.02 (0.51–1.53) by a one-step weighted fit; a damped iterated two-way fit gave 0.93 (0.09–1.48) b_{P}, with 1,962 of 2,000 bootstrap draws converging, and its one-step counterpart 1.08 (0.47–1.66). The undamped two-way iteration oscillated between two values and did not converge. A week fixed-effects fit gave 1.42 (0.96–1.89) b_{P}.

The pooled and state fixed-effects fits can also be compared through their crossover counts. The cross-count ratio, pooled b/a divided by state fixed-effects b over mean state intercept, was 2.16 (0.99–7.64). It factorises exactly into a slope ratio, pooled b over state fixed-effects b, of 1.48 (0.99–3.66), and an intercept ratio, mean state intercept over pooled a, of 1.47 (0.99–2.29); the slope ratio accounts for half of the log cross-count ratio (0.50, 0.31–0.74) (Table S9). Under single-week resampling all three intervals include 1; under two-week blocks all three still do, and under three-week blocks the three lower limits are 1.01–1.03. In the audit, generator S4, in which all states share one counting process, gave a pooled-to-within slope ratio of 1.14 (Monte Carlo range 0.99–1.37) (Table 4). Structured heterogeneity alone thus raises the pooled slope above the within-state slope. The observed slope ratio lies above the S4 value, but its interval overlaps the S4 range, so the comparison does not establish between-state differences in counting.

The counting slope also does not identify reporting probability or offspring dispersion. With independent transmission heterogeneity of the S1 kind, the expected b/b_{P} ranged from 0.37 (Poisson offspring, $\rho = 1$) to 2.89 ($k = 0.1$, $\rho = 1$) across four reporting probabilities and five offspring laws (Fig. 4; Table S10). The observed interval contains the expected value in 10 of the 20 cells: all five offspring laws at $\rho = 0.1$, $k = 1$ at every reporting probability and $k = 0.3$ at $\rho = 0.25$ and 0.5. Very different combinations of under-reporting and superspreading are compatible with the same slope, and heterogeneity of the S4 kind would lower the expected slopes (S4 0.73 against S1 0.85 in Table 4).

5.3 Dispersion-to-counting crossover count

The dispersion-to-counting crossover count was $b/a = 37.4$ admissions per week (18.4–72.4) for R_A. The point estimate was 49.7 (25.0–96.6) when rows required $C_t \geq 10$ (1,535 rows), 52.8 (19.0–122.9) at $C_t \geq 20$ (1,203 rows) and 53.1 (-8.4 to 180.1) at $C_t \geq 50$ (776 rows), where the ratio was nonpositive in 4.8% of draws. A continuity correction of 1.0 instead of 0.5 gave 33.1 (16.3–66.3) and a window threshold of 1.10 gave 38.6 (20.6–70.3) on 1,564 rows (Table 5; Fig. 5A). The value 37.4 is an in-sample descriptive crossover for this panel and window rule, not a sample-size rule for surveillance.

Fitted separately, the four seasons gave 50.9, 113.7, 45.9 and 15.7 on 365, 499, 356 and 528 rows. These values are descriptive, and each has a wide interval, from 4.1–71.6 in 2025–26 to 30.2–315.1 in 2023–24. A model with season-specific intercepts and a common slope ($b = 0.607$, 0.415–0.794) gave season crossover counts of 38.3, 85.7, 39.5 and 33.7 (Table S11). With a common slope, equal season intercepts are equivalent to equal season crossover counts. Equality was not rejected at the 5% level. The pre-specified primary specification, a Wald test on the log intercepts (accounting for inter-season bootstrap covariance), gave $W = 5.52$ on 3 degrees of freedom ($p = 0.138$). Cochran's Q, evaluated

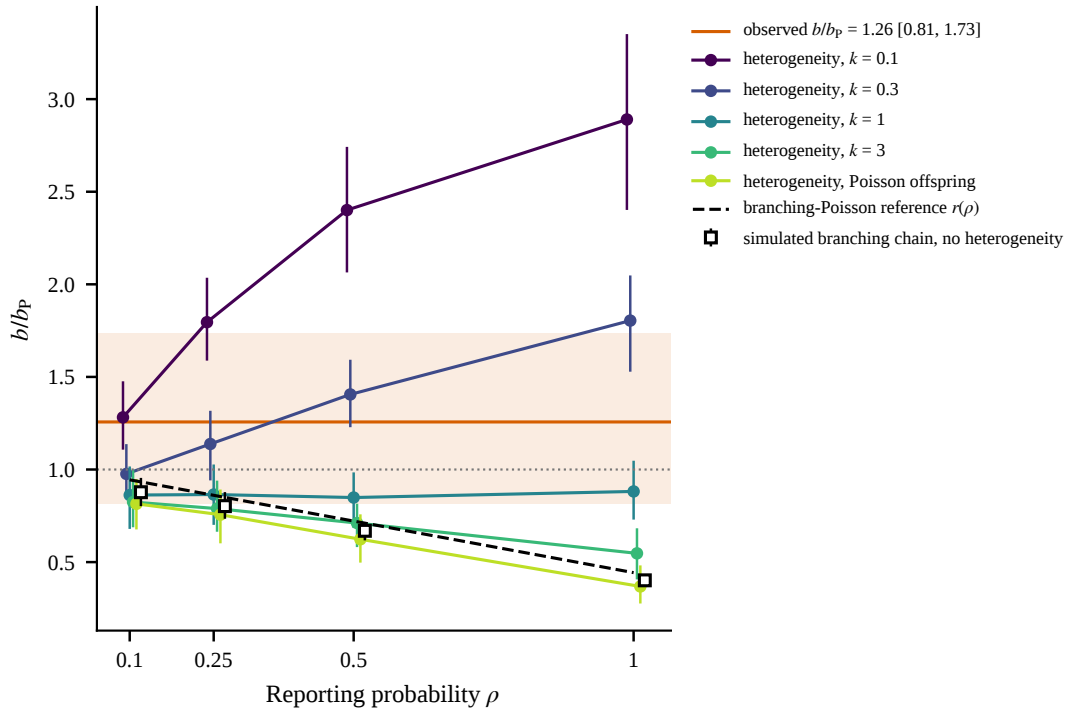


Figure 4 Expected counting slope relative to the independent-Poisson reference, b/b_P, against reporting probability ρ for five offspring laws. Every simulated panel includes independent transmission heterogeneity as in generator S1 and is count-aligned as in the audit; points are means over 100 synthetic panels on the real R_A design and bars are Monte Carlo 2.5–97.5% ranges. The Poisson-offspring curve declines with ρ because, given the realised parent count, the denominator carries only the binomial reporting noise $(1-\rho)/C_t$, as in the branching-Poisson reference. The dashed line is the analytic branching-Poisson reference $r(\rho)$ of Eq. (17) and open squares are simulated branching chains with Poisson offspring and no heterogeneity (Table S2). The horizontal line and shaded band mark the observed b/b_P and its 95% confidence interval.

Table 5 Sensitivity of the pooled loss model to filters, continuity correction, window rule and season (bootstrap 95% confidence interval in brackets)

Specification	n	a	b	b/a	b/b_P
Primary (C_t, $D_t \geq 5$; cc 0.5; growth rule 1.05)	1748	0.0150 [0.0101, 0.0206]	0.560 [0.362, 0.769]	37.4 [18.4, 72.4]	1.26 [0.81, 1.73]
Counts ≥ 10	1535	0.0136 [0.0090, 0.0188]	0.677 [0.436, 0.938]	49.7 [25.0, 96.6]	1.54 [0.99, 2.14]
Counts ≥ 20	1203	0.0133 [0.0081, 0.0189]	0.701 [0.332, 1.092]	52.8 [19.0, 122.9]	1.63 [0.76, 2.52]
Counts ≥ 50	776	0.0129 [0.0073, 0.0191]	0.686 [-0.155, 1.479]	53.1 [-8.4, 180.1] (b/a ≤ 0 in 4.8% of draws)	1.64 [-0.38, 3.59]
Continuity correction 1.0	1748	0.0154 [0.0102, 0.0214]	0.510 [0.326, 0.728]	33.1 [16.3, 66.3]	1.15 [0.74, 1.64]
Growth-window rule 1.10	1564	0.0141 [0.0097, 0.0187]	0.543 [0.357, 0.760]	38.6 [20.6, 70.3]	1.26 [0.83, 1.75]
Season 2022-23	365	0.0137 [0.0067, 0.0210]	0.697 [0.350, 1.083]	50.9 [19.8, 134.8] (5 draws not converged)	1.62 [0.80, 2.50]
Season 2023-24	499	0.0062 [0.0025, 0.0141]	0.702 [0.403, 0.963]	113.7 [30.2, 315.1] (2 draws not converged)	1.51 [0.85, 2.10]
Season 2024-25	356	0.0127 [0.0058, 0.0232]	0.583 [0.161, 1.049]	45.9 [8.2, 147.2]	1.38 [0.36, 2.50]
Season 2025-26	528	0.0225 [0.0085, 0.0355]	0.353 [0.129, 0.718]	15.7 [4.1, 71.6]	0.78 [0.28, 1.61]

Each row refits the affine FGLS on the subset or variant with a state-week bootstrap; the primary row is the primary bootstrap (2,000 draws, identical to Table S1), the other rows use 2,000 draws each. Where b/a ≤ 0 (b ≤ 0) in more than 2.5% of draws its percentile interval is unstable; draws whose FGLS fit did not converge are dropped and counted.

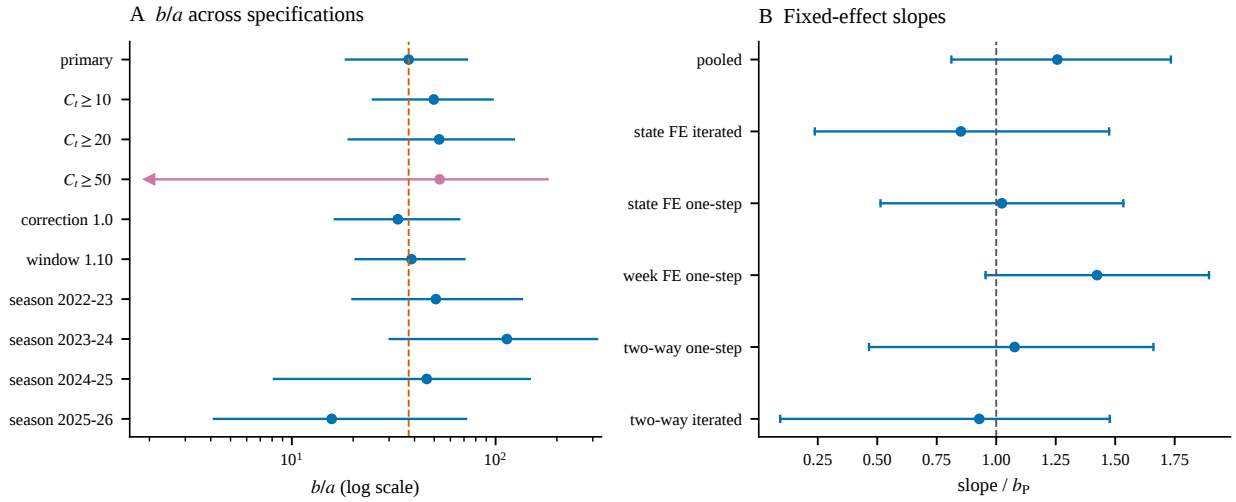


Figure 5 Sensitivity of the dispersion summaries for R_A. (A) Dispersion-to-counting crossover count b/a (log scale) under the primary specification, minimum-count filters, a continuity correction of 1.0, a window threshold of 1.10 and separate season fits; the dashed line marks the primary estimate. For $C_t \geq 50$ the ratio was nonpositive in more than 2.5% of draws, so the lower bound lies below the axis. (B) Counting slopes relative to the independent-Poisson reference for pooled, state, week and two-way fixed-effects fits; the dashed line marks 1. Bars are bootstrap 95% confidence intervals.

using diagonal inverse-variance weights without off-diagonal covariance, was 4.57 ($p = 0.206$, $I^2 = 34\%$). On the raw intercept scale the Wald test gave $p = 0.064$, and a test on separate season fits gave $p = 0.220$. These tests have limited power. Reaching 80% power at the 5% level along the observed pattern requires the deviations to be 1.41 times larger, corresponding to a pattern-specific minimum detectable ratio of 3.71 between the highest and lowest season intercepts; the observed ratio is 2.54. The second season (2023–24) had a lower fitted intercept (0.0071 vs ∼0.016 in the others), pushing its crossover count higher (85.7), but finite-sample power precludes treating this as definitive biological divergence. Even when differences are detected, the between-season variance itself can be estimated only to a relative SE of at least 0.82 with four seasons (Appendix B.3).

The interval choice has almost no effect on the ratio. Replacing the primary interval with a mean of 3.2 days changed b/a by 0.17 (-0.65 to 0.89), and the intrinsic generation-time estimate with standard deviation 2.1 days changed it by 0.49 (-2.55 to 3.04), with b/b_P 1.27 and 1.23 (Table S13). The interval rescales a and b together, so it affects the levels of the two summaries but not their ratio.

5.4 Shape of the dispersion curve

Held-out comparison of the affine model with a power law $\lambda C^{-\gamma}$ did not separate the two forms. Leaving out one state at a time, the power-minus-affine score difference was -0.0020 (paired standard error 0.0068), with 26 of 51 folds favouring the affine model; leaving out one season at a time it was 0.0046 (0.0087), with 3 of 4 folds favouring the affine model (Table S14; Fig. S2). The affine form is used here as a working summary, not as an established law for surveillance dispersion.

6 Discussion

The theoretical part of the paper makes one distinction precise. With fixed, known reporting, exogenous exposure and conditionally independent counting errors, the risk for the current value is A/m and the risk for the realised mean of the observed periods is $A/(Tm)$. The process mean adds the variance of the realised mean, which more cases within the same periods do not reduce. The count at which counting error equals between-period variance is the half-weight point of the corresponding shrinkage estimator, as in small-area estimation and disease mapping [14, 15]. It belongs with its estimator, target and sampling experiment and is not a property of a pathogen. The efficiency statement holds within a class of conditionally unbiased estimators with a known denominator. Information from other features of the epidemic curve [6, 35], independent reporting measurements or prior structure changes what can be learned, and estimators that accept bias in exchange for lower variance lie outside that class. Appendix B gives the corresponding forecast weight, the allocation of a fixed budget between periods and cases, and a lower bound on the precision with which a few seasons can estimate between-season variation.

The empirical part shows what can and cannot be read from a cross-sectional dispersion curve. Its intercept is a count-independent relative dispersion, the part of the variation in the statistic that does not scale with the count. With reporting fixed across weeks it is the relative variance of transmission across units and weeks; with week-to-week reporting shocks it

also contains their contribution, and single-period moments cannot split the two (Proposition 2). For influenza admissions it is about 0.015. Under fixed reporting this is the variation that a current-value analysis wants to resolve and that a process-mean analysis treats as sampling variation; under volatile reporting it is an upper bound on the error floor of the current value. The counting slope, its ratio to the independent-Poisson or branching-Poisson reference and the thinning response cannot distinguish overdispersed offspring, under-reporting and heterogeneous transmission. The audit produces slopes near the observed one under several combinations, and the grid over ρ and k admits values from heavy under-reporting with strong superspreading to near-complete reporting with moderate dispersion. The dispersion-to-counting crossover count was about 37 admissions per week for the pooled panel; season estimates differed descriptively, and a test of equal season intercepts did not detect a difference. The crossover marks where counting noise stops dominating the cross-sectional dispersion of this statistic, not where additional cases stop being useful for a particular target.

6.1 Real-time estimation

All analyses used a retrospective snapshot in which counts are close to final. In real time, reporting delays leave the most recent weeks incomplete, and right-censoring biases the latest counts downwards in proportion to the delay distribution [36, 37]. The current-value target is affected first and most: its risk is dominated by the last one or two periods, exactly those that are incomplete, whereas a realised or process mean over many periods dilutes them. The R_A statistic also needs C_{t+1}, so it is available only one week after its reference week, and estimates near the end of a series combine smaller effective counts with nowcasting uncertainty. Real-time renewal methods address this by modelling delays and the infection process jointly [2, 38, 39]. In the present framework, nowcasting adds a correction-error term to the counting term, and its variance grows as the delay distribution places more mass beyond the current week. A crossover count estimated from retrospective data will therefore overstate the precision available in real time.

Estimation windows raise a related issue. Widening a window in renewal estimators accumulates incidence while imposing constancy of the reproduction number over a longer interval [2, 3]. This trades temporal resolution for precision and is not equivalent to observing more independent periods; the averages in Table 1 instead give each period its own transmission value. Applying their formulas to overlapping windows requires explicit treatment of the overlap.

6.2 Reporting that depends on transmission

Proposition 2 assumes that the reporting probability is independent of transmission. In practice testing, care seeking and admission thresholds can change with epidemic activity [40, 41]. Write $z_t = p_t/\rho$ for the relative reporting probability. Given R_t, the estimator then has conditional bias

$$\mathrm{E}(\hat{R}_t \mid R_t) - R_t = [\mathrm{E}(z_t \mid R_t) - 1]\, R_t. \tag{19}$$

By Eq. (19), if reporting rises with transmission, so that $\mathrm{Cov}(z_t, R_t) > 0$, high-transmission periods are over-counted and low-transmission periods under-counted, the estimated swings exaggerate the true ones, and the fitted intercept overstates the relative dispersion of transmission. If reporting falls with transmission, for example when admission capacity saturates or testing is rationed at the peak, the estimates damp the swings and the intercept understates it. In both cases the process-mean estimate is biased by $\mathrm{Cov}(z_t, R_t)$, and the bias does not shrink with more cases. Neither direction can be detected from the counts alone; it requires external information on testing or admission practice. The participation adjustment used here addresses only facility participation, and it changed the intercept negligibly.

6.3 Choice of interval distribution

The transform in Eq. (15) needs an interval distribution. We kept the household serial interval of Cowling et al. [31] as the primary choice because it is directly estimated for influenza, widely used and fixed before the analyses reported here. The intrinsic generation-time estimate of Chan et al. [33], the more appropriate quantity for a renewal mapping, served as a sensitivity analysis. Serial intervals, intrinsic generation intervals and realised household generation intervals are different quantities [32], and the mapping from growth rate to reproduction number depends on which is used [30]. The levels of a and b change with the interval, but their ratio and the slope ratio to the independent-Poisson reference change by less than their sampling uncertainty, so the conclusions about what the diagnostics measure do not depend on this choice.

6.4 Identification and limitations

Separating the mechanisms behind the counting slope requires information that a single cross-section lacks. Temporal structure can separate persistent transmission variation from independent counting and reporting noise under explicit state-space or renewal models [24, 25, 42], and external measurements of reporting can constrain ρ and its variation. The audit adds a further reason for caution, through the shape of the curve. The observed affine-minus-power contrast lay at or below the 1% quantile of every generator with constant transmission heterogeneity (S1 and S3–S6). A generator in which heterogeneity shrinks with the count scale (S7a) matched the intercept by construction and placed the observed contrast

at its 43% quantile, so the observed curve is consistent with scale-dependent heterogeneity, in which larger jurisdictions vary less in relative terms. The same generator gave a crossover count of 51.4 (Monte Carlo range 39.5–65.9), above the observed 37.4; combining moderate scale-dependent heterogeneity with period-level reporting volatility (S8 Combo), however, yields an intercept of 0.0148, a crossover count of 41.7 (30.6–52.1, covering the observed 37.4), a slope ratio of 1.298, and an observed contrast at the 8.0% quantile. Even with combinations that accommodate all three summaries, observational count diagnostics cannot uniquely identify the underlying mechanism without longitudinal or auxiliary data; reporting that varies with jurisdiction size could leave a similar trace.

Several limitations apply. The growth windows were early-growth segments of four seasons, inherited from the original analysis pipeline and not tuned here. Inference conditions on that selection; a stricter window threshold of 1.10 gave a crossover count of 38.6, and four seasons do not support statements about other epidemic regimes. The analyses used a retrospective snapshot, so reporting delays and right-censoring of recent weeks were not modelled (Section 6.1). The ratio statistic shares C_t between adjacent weeks, so reporting errors in adjacent rows are negatively correlated (about $-1/2$ in S6), which the row-level resampling does not represent. The audit used 200 panels per generator, so quantiles of observed values are resolved only to 0.005. Its count alignment matched the median and lower tail of the observed counts but not the upper tail (90th percentile 263–324 against 389). The (ρ, k) grid used independent heterogeneity only, and the G statistic was not audited. Bootstrap draws in which an iterated fit did not converge were dropped and their number reported, which conditions those intervals on convergence. Within-unit slopes were imprecise and sensitive to the weighting scheme. Crossed resampling and block sensitivity address dependence between states in the same week and within states over time under their own assumptions [34, 43]. Facility participation measures only one aspect of ascertainment, and comparisons across reporting regimes are descriptive.

7 Conclusion

The inferential target decides what additional cases buy: more cases shrink the counting error of the current value and of the realised mean of the observed periods, but they do not remove the between-period variation that limits estimation of the process mean. When reporting fluctuates from week to week, that fluctuation leaves an error floor on the current value, and single-period second moments cannot separate it from transmission variation. Cross-sectional dispersion diagnostics therefore measure a count-independent relative dispersion and the counting noise around it rather than a precision limit or a threshold for allocating surveillance effort, and separating under-reporting, superspreading and structured transmission requires temporal models or external data on reporting.

Data and code availability

The analyses used the dataset Weekly Hospital Respiratory Data (HRD) Metrics by Jurisdiction, National Healthcare Safety Network, published by the U.S. Centers for Disease Control and Prevention under the identifier ua7e-t2fy [27], in the snapshot retrieved on 20 October 2026. The data pipeline and fixed analysis windows are available in the public GitHub repository reproduction-number-targets-surveillance. Code for the known-truth audit, the exact thinning calculation and the sensitivity analyses, together with simulation settings, result files and a SHA-256 manifest of inputs and generated assets, is provided as supplementary material. No individual-level or restricted data were used.

Declarations

CRediT authorship contribution statement **Conceptualization:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Software & Validation:** Software, Data curation, Visualization, Validation. **Supervision:** Project administration.

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A Proofs

A.1 Count law and conditional information

The probability generating function of one negative binomial offspring count is $[1 + (R_t/k)(1 - z)]^{-k}$. Conditional independence raises this function to the power n, and substituting $z \mapsto 1 - \rho + \rho z$ for independent reporting gives $[1 + (\rho R_t/k)(1 - z)]^{-nk}$, which proves Eq. (2); its derivatives give the stated moments.

For fixed, known (n, ρ, k) and $R > 0$, the count likelihood has score and Fisher information

$$\dot{\ell}(R) = \frac{c}{R} - \frac{(c + nk)\rho}{k + \rho R} = \mathcal{I}(R)(\hat{R} - R), \qquad \mathcal{I}(R) = \frac{m}{R + \rho R^2/k}. \tag{A.1}$$

By Eq. (A.1), $\mathrm{Var}(\hat{R} \mid R) = \mathcal{I}(R)^{-1}$. The support does not depend on R, so the regular one-parameter Cramér–Rao inequality applies to estimators unbiased for every R and gives the attained bound stated in Section 2.2 [44]. The bound therefore applies to estimators unbiased for every R with ρ and k known; biased estimators and unknown nuisance parameters require a separate analysis.

A.2 Proof of Proposition 1

Let $e_t = \hat{R}_t - R_t$. The error is conditionally centred, $\mathrm{E}(e_t \mid R_t) = 0$, and its second moments are as follows:

$$\mathrm{E}(e_t^2) = \frac{\mathrm{E}\, R_t + \rho\, \mathrm{E}\, R_t^2/k}{m} = \frac{A}{m}, \qquad \mathrm{E}[e_t(R_t - \overline{R})] = 0, \tag{A.2}$$

Squaring Eq. (4) proves Eq. (3). Any estimator based on the same count and unbiased conditional on every R_t has expected conditional variance at least $\mathrm{E}[\mathcal{I}(R_t)^{-1}] = A/m$, and its process-mean risk adds the same v_R. At fixed ρ, k and transmission law, $[V(m) - V(2m)]/V(m) = 1/[2(1 + m/m_\times)]$ for $v_R > 0$. The comparison changes exposure n; it does not hold unchanged when the reporting probability changes. The shrinkage weight in Eq. (5) is $\mathrm{Cov}(R_t, \hat{R}_t)/\mathrm{Var}(\hat{R}_t) = v_R/V(m)$.

A.3 Proof of Corollary 1

Conditional on the path $R_{1:T}$, the e_t are independent and centred, hence pairwise uncorrelated and orthogonal to every square-integrable function of the path. Thus the average of $e_1, \ldots, e_T$ has second moment $A/(Tm)$, and $\mathrm{Var}(\overline{\hat{R}}_T) = \mathrm{Var}(\overline{R}_T) + A/(Tm)$. Under the stated stationary autocovariance, summing all covariance pairs gives $T^2\,\mathrm{Var}(\overline{R}_T) = v_R S(\phi, T)$ with the first expression in Eq. (7), and the finite geometric sum yields the second [45]. Summing spatial covariance pairs in the same way gives Eq. (8), for any admissible correlation matrix.

A.4 Proof of Proposition 2 and Corollary 2

Write $z_t = p_t/\rho$ and $u_t = C_t/m - z_t R_t$. Conditional on (R_t, p_t), u_t has zero mean. Independence of p_t and R_t, with $\mathrm{E}\, z_t = 1$ and $\mathrm{E}\, z_t^2 = 1 + \tau^2$, gives

$$\mathrm{E}(u_t^2) = \frac{\overline{R} + \rho(1 + \tau^2)(\overline{R}^2 + v_R)/k}{m} = \frac{A_H}{m}. \tag{A.3}$$

The error $\hat{R}_t - R_t = u_t + (z_t - 1)R_t$ has uncorrelated terms, and the second has expected square $\tau^2(\overline{R}^2 + v_R) = v_H$, which proves Eq. (11). Because $\mathrm{E}(\hat{R}_t - R_t \mid R_t) = 0$, process-mean risk adds v_R. Under the additional cross-period assumptions of Corollary 2, the errors are centred and independent given the transmission path after integrating out reporting, which gives Eq. (12).

For the identification statement, set $U = (1 + \tau^2)(\overline{R}^2 + v_R)$. The two moments of the scaled count are

$$\mathrm{E}\, \hat{R}_t = \overline{R}, \qquad \mathrm{Var}(\hat{R}_t) = U - \overline{R}^2 + \frac{\overline{R} + \rho U/k}{m}. \tag{A.4}$$

Whenever distinct admissible values of τ^2 give $v_R = U/(1 + \tau^2) - \overline{R}^2 \geq 0$, the moments in Eq. (A.4) coincide, and varying an exogenous m does not separate them. This proves nonidentification by these moments only; it does not assert equality of full count distributions.

For $v_R > 0$, the crossover scale with volatile reporting, $m_{\times,H} = A_H/(v_R + v_H)$, satisfies

$$m_\times - m_{\times,H} = \frac{\tau^2(\overline{R}^2 + v_R)\overline{R}(1 + \rho\overline{R}/k)}{v_R(v_R + v_H)} \geq 0, \tag{A.5}$$

with equality exactly when $\tau = 0$.

A.5 Observed denominator

Let $M_t \sim \text{Bin}(n, \rho)$ be independent of the offspring given n, and write $M_t = m(1+u_t)$, so that $\text{E}\, u_t = 0$ and $\text{Var}(u_t) = (1-\rho)/m$. Then $\tilde{R}_t = \hat{R}_t/(1+u_t)$. To first order in u_t and $\hat{R}_t - R_t$, the error is approximately:

$$\tilde{R}_t - R_t \approx (\hat{R}_t - R_t) - R_t u_t. \tag{A.6}$$

Given (n, R_t), u_t is independent of $\hat{R}_t - R_t$ and centred, so the cross term vanishes and $\text{E}[(\tilde{R}_t - R_t)^2] \approx A/m + (1-\rho)\,\text{E}(R_t^2)/m$, which is Eq. (9). The Monte Carlo check in Table 2 draws R_t from a gamma law with the stated moments and would discard draws with $M_t = 0$, none of which occurred at the scales shown. The approximation neglects the convexity of $1/(1+u_t)$, which inflates the error when m is small; at $m = 10$ the Monte Carlo error is 1.64 times the delta value.

A.6 Reporting shocks in the ratio statistic

Let the count of week t in a state be reported with probability ρz_t, where the shocks z_t are independent between weeks with $\log z_t \sim N(-s^2/2, s^2)$. To first order, the statistic in Eq. (15) satisfies $\log R_{A,t} \approx \text{const} + \kappa(g)\log g_t$ with $\kappa(g) = (g_\mu/7)/[1+\theta\log(g)/7]$, because $\alpha\theta = g_\mu$. The shocks enter the observed growth factor as $\log g_t = \log g_t^{\text{tr}} + \eta_t$, with $\eta_t = \log z_{t+1} - \log z_t$ and g_t^{tr} the growth factor without shocks. Hence $\text{Var}(\eta_t) = 2s^2$, and the relative variance that the shocks add to the ratio statistic is

$$\tilde{v}_H \approx 2\kappa^2 s^2, \tag{A.7}$$

which does not scale with count and therefore enters the intercept a. A reporting level that is constant over consecutive weeks of a state ($z_{t+1} = z_t$) gives $\eta_t = 0$ and cancels. Consecutive rows share one shock with opposite signs, so $\text{Cov}(\eta_t, \eta_{t+1}) = -s^2$ and their correlation is $-1/2$; row-level resampling treats these errors as independent. For the realised mean over T consecutive rows of one state the mean of the shock terms telescopes to $(\log z_{T+1} - \log z_1)/T$, so their contribution to the error of the realised mean falls as T^{-2} rather than T^{-1}. With κ near 0.5 for the primary interval and $s = 0.122$ as in generator S6, Eq. (A.7) gives a shock component of about half the observed intercept.

B Extensions

B.1 Forecasting and shrinkage

Under fixed, known reporting and the stationary covariance of Corollary 1, with the counting error at t centred given the transmission path, the best forecast of R_{t+1} of the form $\overline{R} + \lambda(\hat{R}_t - \overline{R})$ has

$$\lambda^* = \frac{\phi v_R}{v_R + A/m} = \phi\,\gamma(m), \qquad \text{MSE}_{\text{lin}} = v_R - \frac{\phi^2 v_R^2}{v_R + A/m}, \tag{B.1}$$

which follows from $\text{Cov}(\hat{R}_t, R_{t+1}) = \phi v_R$. The forecast weight in Eq. (B.1) is the shrinkage weight discounted by persistence, and its limit risk is $(1-\phi^2)v_R$. For the process mean, shrinking $\hat{R}_t$ towards a fixed benchmark μ_0 with oracle weight $V(m)/[V(m) + (\mu_0 - \overline{R})^2]$ can give risk below v_R; this lies outside the conditionally unbiased class of Proposition 1, and estimating the weight requires its own risk analysis.

B.2 Allocation between periods, cases and coverage

Proposition 3 (Budget allocation between periods and cases)**.** *Under H1–H4, independent, identically distributed transmission periods, conditionally independent counts, fixed ρ and linear costs, let the budget be $\mathcal{B} = T(c_0 + c_1 m)$ with $c_0, c_1 > 0$, and treat $T \geq 1$ and $m > 0$ as continuous. If $v_R > 0$ and $\mathcal{B} > c_0$, process-mean risk $(c_0 + c_1 m)(v_R + A/m)/\mathcal{B}$ is strictly convex in m and is minimised at*

$$m^* = \min\{(c_0 m_\times/c_1)^{1/2},\ (\mathcal{B} - c_0)/c_1\}. \tag{B.2}$$

The optimum in Eq. (B.2) has the form of cluster-size allocation in survey sampling and multilevel design [12, 46]: cases per period should grow only with the square root of $m_\times$ and of the relative cost of a period. Integer designs compare the risk over feasible integer T with $n_T = \lfloor(\mathcal{B}/T - c_0)/(c_1\rho)\rfloor$. Under temporal dependence the variance term becomes $v_R S(\phi, T)/T^2$ and no closed-form optimum follows.

Corollary 3 (Coverage against completeness)**.** *Under H2–H5, let p_t be the participating fraction of ν independent reporting units, each participating with probability ρ, so that $\tau^2 = (1-\rho)/(\rho\nu)$. A proportional increase in completeness ρ reduces single-period process-mean risk more than the same proportional increase in exposure n exactly when $m > \rho^2(\nu - 1)/k$.*

Write the single-period risk as $v_R + v_H + A_H/m$ with $v_H = \tau^2 Q$, $Q = \overline{R}^2 + v_R$ and $m = \rho n$, and differentiate with respect to $\log \rho$ and $\log n$. The difference of the two derivatives is

$$\frac{\partial \,\text{risk}}{\partial \log \rho} - \frac{\partial \,\text{risk}}{\partial \log n} = -\frac{Q}{kn}\left[\frac{kn+\rho}{\rho\nu} - 1\right]. \tag{B.3}$$

Because $Q/(kn) > 0$, the difference is negative, and a proportional increase in ρ lowers the risk more than the same increase in n, exactly when $kn + \rho > \rho\nu$, that is, when $m > \rho^2(\nu - 1)/k$. The mechanism is assumed; observed facility participation alone does not imply it.

B.3 Number of seasons

Suppose $S \geq 2$ season effects are independent Gaussian with unknown mean and variance $v_S > 0$, and the count law given the effects does not depend on these parameters. Any estimator unbiased for v_S then has relative standard error at least $(2/(S-1))^{1/2}$. Observing the effects themselves gives at least as much information as the counts, and their sample variance is the minimum-variance unbiased estimator [11, 44]. With the four seasons analysed here the bound is $(2/3)^{1/2} = 0.82$, and additional admissions within the same seasons leave it unchanged. This bound concerns the precision with which the between-season variance component can be estimated; it is distinct from the power of a test of equal season intercepts, which depends on the size of the season differences relative to their standard errors and is reported in Section 5.3.

C Supplementary tables and figures

Table S1 Pooled affine loss fits and generation-interval sensitivities (bootstrap 95% confidence interval in brackets)

Statistic	Parameter	n	Estimate [95% CI]	Finite/B
R_A	a	1748	0.01495 [0.01006, 0.02063]	2000/2000
R_A	b	1748	0.56 [0.362, 0.7694]	2000/2000
R_A	b/a	1748	37.44 [18.41, 72.37]	2000/2000
G	a	1756	0.08401 [0.05023, 0.1243]	2000/2000
G	b	1756	1.971 [0.9495, 3.06]	2000/2000
G	b/a	1756	23.46 [8.046, 56.36]	2000/2000
R_A (mean 3.2 d)	a	1748	0.01191 [0.007987, 0.01641]	2000/2000
R_A (mean 3.2 d)	b	1748	0.4482 [0.2924, 0.6142]	2000/2000
R_A (mean 3.2 d)	b/a	1748	37.62 [18.72, 72.05]	2000/2000
R_A (3.2/2.1 d)	a	1748	0.01106 [0.007437, 0.01509]	2000/2000
R_A (3.2/2.1 d)	b	1748	0.4196 [0.2793, 0.5704]	2000/2000
R_A (3.2/2.1 d)	b/a	1748	37.93 [19.5, 71.31]	2000/2000

Pooled rows: primary bootstrap (2,000 draws); the same state-week weights are used for every statistic and generation interval. G rows are descriptive and not audited: the known-truth audit (Section 4.3) covers R_A only. Signed percentile intervals; ratios are defined only for positive denominators. Finite/B gives the number of finite bootstrap draws for each estimate.

Table S2 Branching-Poisson and independent-Poisson slope references: analytic, simulated and observed

Reference	Analytic reference / b_P	Simulated b/b_P (no heterogeneity)	Observed b / reference
Branching-Poisson, $\rho = 0.1$	0.944	0.877 [0.802, 0.955]	1.33 [0.86, 1.84]
Branching-Poisson, $\rho = 0.25$	0.861	0.802 [0.734, 0.878]	1.46 [0.94, 2.02]
Branching-Poisson, $\rho = 0.5$	0.722	0.669 [0.618, 0.714]	1.74 [1.12, 2.41]
Branching-Poisson, $\rho = 1$	0.443	0.402 [0.373, 0.434]	2.84 [1.83, 3.94]
Independent-Poisson b_P	1	0.930 [0.853, 1.005]	1.26 [0.81, 1.73]

The branching-Poisson reference is $b_{P,br}(\rho)$ = mean over the observed retained rows of κ^2 [(1 - ρ) + (C_t + 0.5)/(C_{t+1} + 0.5)]; independent-Poisson reference b_P = 0.4455. Simulated: pure branching chain with Poisson offspring and binomial reporting ρ (independent-Poisson counts, S0, in the last row), no transmission heterogeneity, count-aligned as in the audit; 200 panels each, Monte Carlo 2.5-97.5% in brackets. Dividing the simulated reference by the S0 recovery factor (0.930, reflecting the common finite-sample downward bias of gamma FGLS) yields normalized references of 0.943, 0.862, 0.720 and 0.432, in exact agreement with the analytic column; against this normalized benchmark the observed slope ratios become 1.24, 1.36, 1.62 and 2.64. Observed: primary bootstrap (2,000 draws), $b_{P,br}$ recomputed with the weights of each draw. The ratio of the observed slope to the reference depends on the reporting probability.

Table S3 Exact expected thinning increment against the delta prediction b_P

q	Exact rate	Exact rate / b_P	Monte Carlo rate [95% CI]	Leverage in c	Positive-part mass (max)
0.8	0.4672	1.049	0.4588 [0.3731, 0.5557]	0.001	0.0e+00
0.6	0.4845	1.087	0.4745 [0.4064, 0.5532]	0.004	0.0e+00
0.5	0.4957	1.113	0.4969 [0.4243, 0.5606]	0.009	0.0e+00
0.4	0.5050	1.133	0.4997 [0.4397, 0.5611]	0.021	0.0e+00
0.3	0.5051	1.134	0.5067 [0.4482, 0.5539]	0.051	0.0e+00
0.2	0.4798	1.077	0.4749 [0.4292, 0.5229]	0.151	0.0e+00
0.1	0.3836	0.861	0.3804 [0.3337, 0.4204]	0.763	0.0e+00
c, all q	0.4084	0.917	0.4051 [0.3656, 0.4413]	1.000	–
c, without q = 0.1	0.4882	1.096	–	–	–

Rate = (b_q - b_1)/(1/q - 1) from the affine FGLS fit of the exact expected thinned loss (binomial supports summed, mass below 1e-13 dropped); b_P = 0.4455 is the delta-method prediction. Leverage: share of the least-squares rate through the origin carried by each q. Check: q = 0.5: independent-thinning MC 0.4903 (SE 0.0048) vs expected-loss 0.4957; q = 0.1: independent-thinning MC 0.3832 (SE 0.0024) vs expected-loss 0.3836.

Table S4 Shared-count binomial thinning of the R_A sample (n=1748): estimates, simultaneous bands and simulation diagnostics

Sample	q / fit	Metric	Estimate [95% CI]	Simultaneous 95%	Finite/B
all rows	q=0.8	b	0.6747 [0.4905, 0.9034]	–	1000/1000
all rows	q=0.8	c	0.4588 [0.3731, 0.5557]	[0.3381, 0.5795]	1000/1000
all rows	q=0.8	Δa	-0.0003183 [-0.000786, 0.00005128]	[-0.0008833, 0.0002467]	1000/1000
all rows	q=0.8	c/b_P	1.03 [0.8476, 1.242]	–	1000/1000
all rows	q=0.6	b	0.8763 [0.6941, 1.103]	–	1000/1000
all rows	q=0.6	c	0.4745 [0.4064, 0.5532]	[0.3751, 0.574]	1000/1000
all rows	q=0.6	Δa	-0.000836 [-0.001963, -0.000006555]	[-0.00213, 0.0004583]	1000/1000
all rows	q=0.6	c/b_P	1.065 [0.915, 1.241]	–	1000/1000
all rows	q=0.5	b	1.057 [0.858, 1.275]	–	1000/1000
all rows	q=0.5	c	0.4969 [0.4243, 0.5606]	[0.4047, 0.589]	1000/1000
all rows	q=0.5	Δa	-0.001318 [-0.002744, -0.00009326]	[-0.003126, 0.0004903]	1000/1000
all rows	q=0.5	c/b_P	1.115 [0.9621, 1.252]	–	1000/1000
all rows	q=0.4	b	1.31 [1.112, 1.534]	–	1000/1000
all rows	q=0.4	c	0.4997 [0.4397, 0.5611]	[0.4158, 0.5837]	1000/1000
all rows	q=0.4	Δa	-0.001794 [-0.00376, -0.0001502]	[-0.004272, 0.0006837]	1000/1000
all rows	q=0.4	c/b_P	1.122 [0.9903, 1.257]	–	1000/1000
all rows	q=0.3	b	1.742 [1.503, 1.958]	–	1000/1000
all rows	q=0.3	c	0.5067 [0.4482, 0.5539]	[0.4326, 0.5808]	1000/1000
all rows	q=0.3	Δa	-0.002454 [-0.004852, -0.000159]	[-0.005728, 0.0008198]	1000/1000
all rows	q=0.3	c/b_P	1.137 [1.008, 1.249]	–	1000/1000
all rows	q=0.2	b	2.46 [2.221, 2.711]	–	1000/1000
all rows	q=0.2	c	0.4749 [0.4292, 0.5229]	[0.4126, 0.5373]	1000/1000
all rows	q=0.2	Δa	-0.002484 [-0.005751, 0.0005387]	[-0.006867, 0.001899]	1000/1000
all rows	q=0.2	c/b_P	1.066 [0.9728, 1.173]	–	1000/1000
all rows	q=0.1	b	3.983 [3.509, 4.379]	–	1000/1000
all rows	q=0.1	c	0.3804 [0.3337, 0.4204]	[0.3234, 0.4373]	1000/1000
all rows	q=0.1	Δa	0.0007312 [-0.00305, 0.005984]	[-0.005595, 0.007057]	1000/1000
all rows	q=0.1	c/b_P	0.8537 [0.7415, 0.9566]	–	1000/1000
all rows	all q	c	0.4051 [0.3656, 0.4413]	–	1000/1000
all rows	all q	c/b_P	0.9093 [0.8101, 0.9995]	–	1000/1000
all rows	all q	Δa rate	-0.0001293 [-0.0006192, 0.0004699]	–	1000/1000
all rows	without $q = 0.1$	c	0.4848 [0.4387, 0.5279]	–	1000/1000
all rows	without $q = 0.1$	c/b_P	1.088 [0.9936, 1.186]	–	1000/1000
all rows	without $q = 0.1$	Δa rate	-0.0008058 [-0.001703, 0.00005179]	–	1000/1000

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Sample	q / fit	Metric	Estimate [95% CI]	Simultaneous 95%	Finite/B
C_t at least 50	q=0.8	b	0.7797 [-0.007567, 1.573]	–	1000/1000
C_t at least 50	q=0.8	c	0.442 [0.2771, 0.6691]	[0.1839, 0.7002]	1000/1000
C_t at least 50	q=0.8	Δa	-0.00007119 [-0.0004223, 0.0001553]	[-0.0004704, 0.000328]	1000/1000
C_t at least 50	q=0.8	c/b_{P}	1.051 [0.6634, 1.606]	–	1000/1000
C_t at least 50	q=0.6	b	0.969 [0.1971, 1.773]	–	1000/1000
C_t at least 50	q=0.6	c	0.4497 [0.3339, 0.6318]	[0.2558, 0.6435]	1000/1000
C_t at least 50	q=0.6	Δa	-0.0001877 [-0.0009664, 0.0002652]	[-0.0009938, 0.0006184]	1000/1000
C_t at least 50	q=0.6	c/b_{P}	1.069 [0.8015, 1.51]	–	1000/1000
C_t at least 50	q=0.5	b	1.126 [0.3593, 1.891]	–	1000/1000
C_t at least 50	q=0.5	c	0.4567 [0.3405, 0.6213]	[0.273, 0.6405]	1000/1000
C_t at least 50	q=0.5	Δa	-0.0002913 [-0.001459, 0.0003529]	[-0.001457, 0.0008742]	1000/1000
C_t at least 50	q=0.5	c/b_{P}	1.086 [0.8187, 1.476]	–	1000/1000
C_t at least 50	q=0.4	b	1.351 [0.6322, 2.134]	–	1000/1000
C_t at least 50	q=0.4	c	0.4548 [0.3568, 0.593]	[0.297, 0.6126]	1000/1000
C_t at least 50	q=0.4	Δa	-0.0004078 [-0.001854, 0.0004131]	[-0.00194, 0.001125]	1000/1000
C_t at least 50	q=0.4	c/b_{P}	1.081 [0.8483, 1.413]	–	1000/1000
C_t at least 50	q=0.3	b	1.723 [1.005, 2.495]	–	1000/1000
C_t at least 50	q=0.3	c	0.4515 [0.3668, 0.5711]	[0.3174, 0.5855]	1000/1000
C_t at least 50	q=0.3	Δa	-0.0005649 [-0.00247, 0.0006027]	[-0.002581, 0.001452]	1000/1000
C_t at least 50	q=0.3	c/b_{P}	1.073 [0.8687, 1.355]	–	1000/1000
C_t at least 50	q=0.2	b	2.522 [1.77, 3.276]	–	1000/1000
C_t at least 50	q=0.2	c	0.4633 [0.3813, 0.5611]	[0.3434, 0.5831]	1000/1000
C_t at least 50	q=0.2	Δa	-0.0009427 [-0.00351, 0.0008025]	[-0.003825, 0.00194]	1000/1000
C_t at least 50	q=0.2	c/b_{P}	1.102 [0.9074, 1.325]	–	1000/1000
C_t at least 50	q=0.1	b	4.97 [4.216, 5.798]	–	999/1000
C_t at least 50	q=0.1	c	0.4778 [0.4127, 0.5552]	[0.3835, 0.5721]	999/1000
C_t at least 50	q=0.1	Δa	-0.001594 [-0.005615, 0.001213]	[-0.006151, 0.002964]	999/1000
C_t at least 50	q=0.1	c/b_{P}	1.136 [0.9843, 1.323]	–	999/1000
C_t at least 50	all q	c	0.4734 [0.409, 0.5503]	–	999/1000
C_t at least 50	all q	c/b_{P}	1.126 [0.9702, 1.316]	–	999/1000
C_t at least 50	all q	Δa rate	-0.0001928 [-0.0006891, 0.0001482]	–	999/1000
C_t at least 50	without $q = 0.1$	c	0.4594 [0.3808, 0.5603]	–	1000/1000
C_t at least 50	without $q = 0.1$	c/b_{P}	1.092 [0.9026, 1.327]	–	1000/1000
C_t at least 50	without $q = 0.1$	Δa rate	-0.0002434 [-0.0009563, 0.0002083]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.8	b	0.5473 [-0.2021, 1.391]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.8	c	0.4994 [0.3073, 0.7827]	[0.1979, 0.8009]	1000/1000
C_t, C_{t+1} at least 50	q=0.8	Δa	-0.0001674 [-0.0006684, 0.0001096]	[-0.0006867, 0.0003519]	1000/1000
C_t, C_{t+1} at least 50	q=0.8	c/b_{P}	1.194 [0.722, 1.896]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.6	b	0.754 [0.02843, 1.586]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.6	c	0.4973 [0.3582, 0.7249]	[0.2631, 0.7315]	1000/1000
C_t, C_{t+1} at least 50	q=0.6	Δa	-0.0004007 [-0.001555, 0.000186]	[-0.001503, 0.0007012]	1000/1000
C_t, C_{t+1} at least 50	q=0.6	c/b_{P}	1.189 [0.854, 1.751]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.5	b	0.9288 [0.1947, 1.748]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.5	c	0.5063 [0.3635, 0.7058]	[0.2898, 0.7229]	1000/1000
C_t, C_{t+1} at least 50	q=0.5	Δa	-0.0006083 [-0.00219, 0.0001859]	[-0.002149, 0.0009321]	1000/1000
C_t, C_{t+1} at least 50	q=0.5	c/b_{P}	1.21 [0.8863, 1.69]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.4	b	1.161 [0.4133, 1.977]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.4	c	0.4921 [0.3785, 0.6457]	[0.3096, 0.6746]	1000/1000
C_t, C_{t+1} at least 50	q=0.4	Δa	-0.00078 [-0.002694, 0.0002525]	[-0.002738, 0.001178]	1000/1000
C_t, C_{t+1} at least 50	q=0.4	c/b_{P}	1.176 [0.9064, 1.548]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.3	b	1.554 [0.8611, 2.363]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.3	c	0.4849 [0.3882, 0.6214]	[0.3327, 0.6371]	1000/1000
C_t, C_{t+1} at least 50	q=0.3	Δa	-0.001058 [-0.003465, 0.0003471]	[-0.00357, 0.001454]	1000/1000
C_t, C_{t+1} at least 50	q=0.3	c/b_{P}	1.159 [0.9271, 1.485]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.2	b	2.382 [1.63, 3.167]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.2	c	0.4899 [0.3955, 0.5995]	[0.3589, 0.6209]	1000/1000

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Sample	q / fit	Metric	Estimate [95% CI]	Simultaneous 95%	Finite/B
C_t, C_{t+1} at least 50	q=0.2	Δa	-0.001588 [-0.004831, 0.0005575]	[-0.005049, 0.001873]	1000/1000
C_t, C_{t+1} at least 50	q=0.2	c/b_{P}	1.171 [0.9456, 1.432]	–	1000/1000
C_t, C_{t+1} at least 50	q=0.1	b	4.863 [4.1, 5.726]	–	999/1000
C_t, C_{t+1} at least 50	q=0.1	c	0.4934 [0.4221, 0.5729]	[0.3956, 0.5913]	999/1000
C_t, C_{t+1} at least 50	q=0.1	Δa	-0.002397 [-0.007015, 0.000824]	[-0.007549, 0.002756]	999/1000
C_t, C_{t+1} at least 50	q=0.1	c/b_{P}	1.18 [1.005, 1.371]	–	999/1000
C_t, C_{t+1} at least 50	all q	c	0.4926 [0.42, 0.5742]	–	999/1000
C_t, C_{t+1} at least 50	all q	c/b_{P}	1.178 [1.004, 1.379]	–	999/1000
C_t, C_{t+1} at least 50	all q	Δa rate	-0.0003058 [-0.000896, 0.00009878]	–	999/1000
C_t, C_{t+1} at least 50	without $q = 0.1$	c	0.4898 [0.3971, 0.6072]	–	1000/1000
C_t, C_{t+1} at least 50	without $q = 0.1$	c/b_{P}	1.171 [0.9533, 1.462]	–	1000/1000
C_t, C_{t+1} at least 50	without $q = 0.1$	Δa rate	-0.0004328 [-0.001331, 0.0001204]	–	1000/1000

Per-q intervals are pointwise; the separate bands are simultaneous over q within each subset. C_t at least 50 requires original current count ≥ 50; C_t, C_{t+1} at least 50 requires both original adjacent counts ≥ 50. c is the slope increment per (1/q-1); b_{P} is the independent-Poisson reference (plug-in delta value). B=1000; R(point)=400; R(inner, actual)=50; precision criterion: MC SE at most 0.1 times the point estimate, applied to the five monitored response summaries; max finite MC SE/|point| over all stored metrics=68.44; max finite MC SE/sampling SE over all stored metrics=1.742; undefined MC ratios=0/219. Point fits 8400/8400; bootstrap fits 1049998/1050000; failed bootstrap averages=2. Any failed inner fit makes that q average undefined; no silent averaging over successful fits. all rows: n=1748; b_{P}=0.4455 [0.4282, 0.4658]. C_t at least 50: n=828; b_{P}=0.4206 [0.4032, 0.4387]. C_t, C_{t+1} at least 50: n=821; b_{P}=0.4183 [0.4008, 0.4359].

Table S5 Small-count variance of the log thinned count relative to the delta approximation

C	q = 0.8	q = 0.6	q = 0.5	q = 0.4	q = 0.3	q = 0.2	q = 0.1
5	1.05	1.11	1.11	1.04	0.86	0.57	0.21
10	1.02	1.05	1.08	1.12	1.13	0.99	0.53
20	1.01	1.02	1.03	1.05	1.09	1.14	0.97
45	1.00	1.01	1.01	1.02	1.03	1.06	1.14
100	1.00	1.00	1.01	1.01	1.01	1.02	1.06
500	1.00	1.00	1.00	1.00	1.00	1.00	1.01

Exact added variance Var log(Bin(C, q) + 0.5) - Var log(C + 0.5) divided by the delta value (1 - q)/(qC). Values below 1 mean that thinning adds less variance than the delta method assumes.

Table S6 Centering stages with a common loss normalizer

Statistic / center	Pooled mean	a / change [95% CI]	b [95% CI]	Finite/B
R_A / pooled	1.206	0.01495 [0.01006, 0.02063]	0.56 [0.362, 0.7694]	2000/2000
R_A / season	1.206	0.01471 [0.009507, 0.01966]	0.5391 [0.3602, 0.7205]	1000/1000
R_A / season linear trend	1.206	0.0154 [0.008972, 0.01901]	0.3936 [0.2717, 0.5763]	1000/1000
R_A / week	1.206	0.009766 [0.007103, 0.01357]	0.4581 [0.3124, 0.5916]	1000/1000
R_A / week, leave own state out	1.206	0.01016 [0.007654, 0.01481]	0.4841 [0.3492, 0.6636]	1000/1000
R_A / week, change in a	1.206	-0.005189 [-0.01015, -0.0006938]	–	1000/1000
R_A / week, reduction in a (%)	1.206	34.7 [5.967, 51.9]	–	1000/1000
G / pooled	1.724	0.08401 [0.05023, 0.1243]	1.971 [0.9495, 3.06]	2000/2000
G / season	1.724	0.08163 [0.0504, 0.1156]	1.842 [0.9591, 2.602]	1000/1000
G / season linear trend	1.724	0.07864 [0.04808, 0.09897]	1.249 [0.6423, 1.988]	1000/1000
G / week	1.724	0.05607 [0.03962, 0.07812]	1.401 [0.7134, 1.912]	1000/1000
G / week, leave own state out	1.724	0.05843 [0.04308, 0.08505]	1.477 [0.8089, 2.138]	1000/1000
G / week, change in a	1.724	-0.02794 [-0.05687, -0.003708]	–	1000/1000
G / week, reduction in a (%)	1.724	33.25 [6.805, 51.63]	–	1000/1000

All five centres use the same pooled mean as the relative-loss denominator within each statistic; the denominator is refitted with weights within each bootstrap draw. The leave-own-state-out week centre removes the focal state from its weekly centre. Reduction is $100(a_{\text{pooled}} - a_{\text{week}})/a_{\text{pooled}}$, in percent; changes are descriptive and not additive variance components. Pooled-centre rows: primary bootstrap (2,000 draws, as in Table S1); the other centres and the week change use this table's own 1000-draw bootstrap. Finite/B is defined as in Table S1.

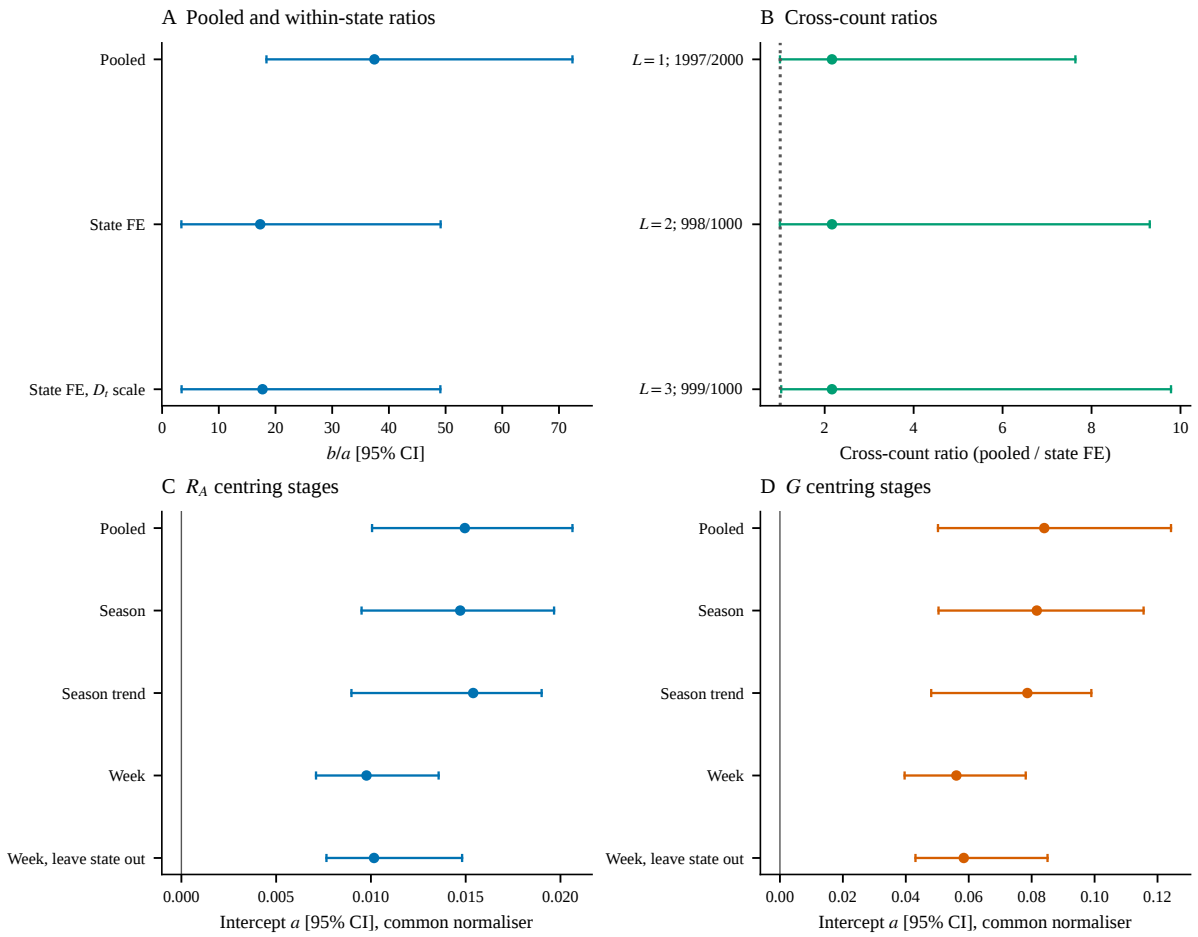


Figure S1 Heterogeneity diagnostics for the R_A and G samples. (A) Crossover count b/a for R_A from the pooled fit, a state fixed-effects fit and a state fixed-effects fit with D_t as count scale. (B) Cross-count ratio, pooled b/a over state fixed-effects b divided by the mean state intercept, under week resampling ($L = 1$) and two- and three-week blocks ($L = 2, 3$), with the number of finite bootstrap draws; the dotted line marks 1. (C, D) Intercept a under five centring choices for R_A and G, each with the pooled mean of its statistic as common normaliser. Bars are bootstrap 95% confidence intervals.

Table S7 Season-specific fits and reporting participation sensitivity

Statistic / period	n	a / change [95% CI]	b / original a	b/a / adjusted a	Finite/B
R_A / 2022-23	365	0.01369 [0.0071, 0.02102]	0.6967	50.91	1000/1000
R_A / 2023-24	499	0.006174 [0.002667, 0.01542]	0.7018	113.7	1000/1000
R_A / 2024-25	356	0.01268 [0.005687, 0.02276]	0.5826	45.93	1000/1000
R_A / 2025-26	528	0.02253 [0.008432, 0.03509]	0.3533	15.68	1000/1000
R_A / reporting adjustment	1748	0.0002939 [-0.0003183, 0.0008476]	0.01495	0.01466	1000/1000
G / 2022-23	367	0.07179 [0.03786, 0.1162]	2.702	37.64	1000/1000
G / 2023-24	503	0.03046 [0.0118, 0.07462]	2.382	78.2	1000/1000
G / 2024-25	357	0.06469 [0.03191, 0.1278]	2.082	32.19	1000/1000
G / 2025-26	529	0.1383 [0.03715, 0.2252]	0.7411	5.359	1000/1000
G / reporting adjustment	1756	0.000373 [-0.003559, 0.003454]	0.08401	0.08364	1000/1000

Season rows report a with its interval and b, b/a as point estimates only. Reporting-adjustment rows use the a column for original-minus-adjusted a [95% CI], b for original a, and b/a for adjusted a; the original normalizer is fixed. Season b and b/a come from separate per-season fits without a stored bootstrap; season b/a with intervals from a common-slope model are given in Table S11. Summer rows are reported in Table S12. Finite/B is defined as in Table S1.

Table S8 Within-state, within-week and two-way slopes against the independent-Poisson reference (bootstrap 95% confidence interval in brackets)

Estimator	Slope [95% CI]	Slope / b_P	Slope / pooled b	Fit status
Pooled affine slope b	0.560 [0.362, 0.769]	1.26 [0.81, 1.73]	–	–
b_P, independent-Poisson reference	0.446 [0.427, 0.465]	–	–	–
State FE, iterated FGLS (b_w)	0.379 [0.107, 0.650]	0.85 [0.24, 1.47]	0.68 [0.26, 1.01]	–
State FE, one-step WLS	0.456 [0.228, 0.682]	1.02 [0.51, 1.53]	0.81 [0.57, 0.96]	single step
Week FE, one-step WLS	0.634 [0.420, 0.846]	1.42 [0.96, 1.89]	1.13 [0.98, 1.34]	single step
State + week FE, one-step WLS	0.480 [0.208, 0.744]	1.08 [0.47, 1.66]	0.86 [0.47, 1.14]	single step
State + week FE, damped iterated FGLS	0.414 [0.040, 0.659]	0.93 [0.09, 1.48]	0.74 [0.09, 1.10]	converged, 124 it.

Primary bootstrap: state-week weights of the primary stream, 2,000 draws, so the pooled row equals Table S1; draws whose iterated fit does not converge are dropped (1962 valid for the damped two-way slope). Undamped two-way iteration: did not converge after 51 iterations (last values 0.4150, 0.4122); damping 0.5 removes the two-cycle. One-step WLS uses the pooled affine working precision; the iterated estimators re-weight with the fixed-effect fitted means.

Table S9 Cross-count ratios (pooled b/a over state fixed-effects b/a), slope ratios and intercept ratios for the R_A sample (n=1748), with week-block sensitivity and convergence

Block	Parameter	Estimate [95% CI]	Finite/B	Nonpositive / diagnostic
L=1	Pooled a	0.01495 [0.01006, 0.02063]	2000/2000	0
L=1	Pooled b	0.56 [0.362, 0.7694]	2000/2000	0
L=1	Pooled b/a	37.44 [18.41, 72.37]	2000/2000	0
L=1	State FE b	0.3794 [0.1069, 0.6502]	2000/2000	0.0015
L=1	State FE mean a	0.02192 [0.01266, 0.03544]	2000/2000	0
L=1	State FE b/mean a	17.3 [3.402, 49.12]	2000/2000	0.0015
L=1	State FE median a	0.01751 [0.01043, 0.02995]	2000/2000	0
L=1	State FE b/median a	21.67 [3.972, 57.83]	2000/2000	0.0015
L=1	State FE row-weighted a	0.02061 [0.01289, 0.03095]	2000/2000	0
L=1	State FE row-weighted ratio	18.41 [3.926, 47.89]	2000/2000	0.0015
L=1	State FE count-weighted a	0.01542 [0.01065, 0.02241]	2000/2000	0
L=1	State FE count-weighted ratio	24.61 [5.617, 55.84]	2000/2000	0.0015
L=1	D_t scale b	0.3691 [0.1053, 0.614]	2000/2000	0.0015
L=1	D_t scale mean a	0.02084 [0.01177, 0.03499]	2000/2000	0
L=1	D_t scale ratio	17.71 [3.438, 49.08]	2000/2000	0.0015
L=1	Slope ratio (pooled b / state FE b)	1.476 [0.9883, 3.662]	1997/2000	0
L=1	Intercept ratio (state FE mean a / pooled a)	1.466 [0.9892, 2.291]	2000/2000	0
L=1	Cross-count ratio (pooled b/a / state FE b/mean a)	2.164 [0.9946, 7.637]	1997/2000	0
L=2	Pooled a	0.01495 [0.01006, 0.02063]	2000/2000	0
L=2	Pooled b	0.56 [0.362, 0.7694]	2000/2000	0
L=2	Pooled b/a	37.44 [18.41, 72.37]	2000/2000	0
L=2	State FE b	0.3794 [0.1045, 0.6646]	1000/1000	0.002
L=2	State FE mean a	0.02192 [0.01245, 0.03616]	1000/1000	0
L=2	State FE b/mean a	17.3 [2.832, 50.14]	1000/1000	0.002
L=2	State FE median a	0.01751 [0.01048, 0.02989]	1000/1000	0
L=2	State FE b/median a	21.67 [3.618, 57.69]	1000/1000	0.002
L=2	State FE row-weighted a	0.02061 [0.01301, 0.0314]	1000/1000	0
L=2	State FE row-weighted ratio	18.41 [3.193, 48.2]	1000/1000	0.002
L=2	State FE count-weighted a	0.01542 [0.01068, 0.02317]	1000/1000	0
L=2	State FE count-weighted ratio	24.61 [4.828, 57.15]	1000/1000	0.002
L=2	D_t scale b	0.3691 [0.09887, 0.6225]	998/1000	0.002
L=2	D_t scale mean a	0.02084 [0.01155, 0.03557]	998/1000	0
L=2	D_t scale ratio	17.71 [2.609, 50.55]	998/1000	0.002
L=2	Slope ratio (pooled b / state FE b)	1.476 [0.9922, 4.29]	998/1000	0
L=2	Intercept ratio (state FE mean a / pooled a)	1.466 [0.9874, 2.291]	1000/1000	0
L=2	Cross-count ratio (pooled b/a / state FE b/mean a)	2.164 [0.9939, 9.309]	998/1000	0
L=3	Pooled a	0.01495 [0.01006, 0.02063]	2000/2000	0
L=3	Pooled b	0.56 [0.362, 0.7694]	2000/2000	0
L=3	Pooled b/a	37.44 [18.41, 72.37]	2000/2000	0
L=3	State FE b	0.3794 [0.09549, 0.6499]	999/1000	0
L=3	State FE mean a	0.02192 [0.01315, 0.03666]	999/1000	0
L=3	State FE b/mean a	17.3 [2.658, 46.48]	999/1000	0
L=3	State FE median a	0.01751 [0.01085, 0.03032]	999/1000	0
L=3	State FE b/median a	21.67 [3.225, 56]	999/1000	0

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(continued)

Block	Parameter	Estimate [95% CI]	Finite/B	Nonpositive / diagnostic
L=3	State FE row-weighted a	0.02061 [0.01343, 0.03143]	999/1000	0
L=3	State FE row-weighted ratio	18.41 [3.027, 45.6]	999/1000	0
L=3	State FE count-weighted a	0.01542 [0.01061, 0.02322]	999/1000	0
L=3	State FE count-weighted ratio	24.61 [4.339, 55.61]	999/1000	0
L=3	D_t scale b	0.3691 [0.08441, 0.6052]	999/1000	0
L=3	D_t scale mean a	0.02084 [0.01186, 0.03658]	999/1000	0
L=3	D_t scale ratio	17.71 [2.365, 47.63]	999/1000	0
L=3	Slope ratio (pooled b / state FE b)	1.476 [1.028, 4.598]	999/1000	0
L=3	Intercept ratio (state FE mean a / pooled a)	1.466 [1.01, 2.365]	999/1000	0
L=3	Cross-count ratio (pooled b/a / state FE b/mean a)	2.164 [1.023, 9.787]	999/1000	0
Point diagnostic	week only	b=0.5258	No CI	gamma-objective fallback
Point diagnostic	state and week	Unavailable	No CI	iteration limit reached

Sample: the R_A specification, n=1748 state-weeks in 51 jurisdictions; the G sample is not used here. Intervals retain signed draws and are conditional on finite estimates when finite/B < 1. Last column gives the nonpositive fraction over all requested draws. FE targets the sampled state occurrences; pooled and FE quantities are computed on the same draw. Cross-count ratio = slope ratio x intercept ratio. L denotes the within-season week-block length. Week-only and state-and-week fits are point diagnostics; failed fits have no estimate or interval. L=1 uses the primary bootstrap (2,000 draws, the weights of Table S1). L=2 and L=3 use 1,000 draws each, and their ratios use the week-block bootstrap. Pooled row: primary bootstrap (2,000 draws). L=1 pooled: 2000 converged, 0 failed. L=1 state: 2000 converged, 0 failed. L=1 state (D_t scale): 2000 converged, 0 failed. L=2 pooled: 1000 converged, 0 failed. L=2 state: 1000 converged, 0 failed. L=2 state (D_t scale): 998 converged, 2 failed. L=3 pooled: 1000 converged, 0 failed. L=3 state: 999 converged, 1 failed. L=3 state (D_t scale): 999 converged, 1 failed.

Table S10 Unthinned slope relative to the independent-Poisson reference over reporting probability and offspring dispersion

Reporting probability ρ	Branching r(ρ)	k = 0.1	k = 0.3	k = 1	k = 3	Poisson offspring
0.1	0.944	1.28 [1.11, 1.47]	0.98 [0.86, 1.13]	0.86 [0.69, 1.01]	0.82 [0.69, 0.99]	0.81 [0.68, 0.94]
0.25	0.861	1.80 [1.59, 2.03]	1.14 [0.95, 1.31]	0.86 [0.71, 1.02]	0.79 [0.67, 0.93]	0.76 [0.61, 0.89]
0.5	0.722	2.40 [2.07, 2.74]	1.41 [1.24, 1.59]	0.85 [0.69, 0.98]	0.71 [0.59, 0.81]	0.62 [0.50, 0.75]
1	0.443	2.89 [2.41, 3.34]	1.80 [1.53, 2.04]	0.88 [0.74, 1.04]	0.55 [0.41, 0.68]	0.37 [0.28, 0.48]

Unthinned b/b_P, with b_P the independent-Poisson reference, under S1 iid heterogeneity (true relative variance = observed a) with negative-binomial offspring of dispersion k and binomial reporting ρ; 100 panels per cell, Monte Carlo 2.5-97.5% in brackets. r(ρ) = $b_{P,br}(\rho)/b_P$ is the analytic branching-Poisson reference on the observed rows (no heterogeneity). Observed b/b_P = 1.26 [0.81, 1.73] (primary bootstrap, 2,000 draws).

Table S11 Season-stratified loss model with a common slope and tests of equal season intercepts (bootstrap 95% confidence interval in brackets)

Quantity	Pooled loss, season intercepts	Season-centred loss, season intercepts
Common slope b	0.607 [0.415, 0.794]	0.586 [0.400, 0.760]
a, 2022-23	0.0159 [0.0087, 0.0269]	0.0162 [0.0085, 0.0263]
a, 2023-24	0.0071 [0.0036, 0.0143]	0.0064 [0.0027, 0.0128]
a, 2024-25	0.0154 [0.0071, 0.0276]	0.0141 [0.0069, 0.0225]
a, 2025-26	0.0180 [0.0075, 0.0278]	0.0185 [0.0057, 0.0277]
b/a, 2022-23	38.3 [17.7, 77.4]	36.2 [17.9, 75.4]
b/a, 2023-24	85.7 [31.8, 193.7]	91.5 [34.5, 228.0]
b/a, 2024-25	39.5 [17.5, 97.6]	41.7 [20.3, 94.6]
b/a, 2025-26	33.7 [17.1, 86.8]	31.7 [17.0, 108.0]
Wald test of equal log a_s (df 3)	W = 5.52, p = 0.138	–
Cochran Q, log a_s (df 3)	Q = 4.57, p = 0.206, I^2 = 34%	–
Wald test of equal a_s (df 3)	W = 7.25, p = 0.064	–
Wald test, separate season fits, log(b_s/a_s) (df 3)	W = 4.41, p = 0.220	–
Minimum detectable highest/lowest a (80% power)	3.71 (observed 2.54)	–

Affine loss with one intercept per season and a common slope (group FGLS); state-week bootstrap, 2,000 draws (1,999 valid). The season-centred column centres x on season means before forming the loss. Tests use the bootstrap covariance of the season parameters (1,995 of 2,000 draws with all parameters finite); with a common slope, equal log a_s is equivalent to equal season b/a. Cochran Q uses diagonal inverse-variance weights without off-diagonal covariance from the common slope. The separate-fit test refits the loss in each season on its own bootstrap stream. Minimum detectable difference: noncentrality 10.90 gives 80% power for a level-0.05 chi-square(3) test; scaling the observed log a_s deviations by f = 1.41 reaches it, which corresponds to the pattern-specific highest/lowest a shown. The highest to lowest season b/a ratio (2.54 [1.52, 6.01]) is descriptive and is not a homogeneity test.

Table S12 Off-season summer descriptors

Period	n	Median count scale	Median participation (%)	IQR(log G)
Summer 2023	772	15.55	94.24	0.4873
Summer 2024	400	17.12	45.59	0.6683
Summer 2025	839	16.25	92.31	0.4882

Descriptive summaries of May-October state-weeks with $D_t \geq 5$: rows, median count scale D_t, median reporting participation (%) and IQR of log G. No model is fitted, so no interval is given. Reporting was voluntary in the 2024 summer only; that contrast is descriptive, not a causal or independently replicated intervention.

Table S13 Generation-interval sensitivity of the pooled loss model (bootstrap 95% confidence interval in brackets)

Generation interval	n	a	b	b/a	b_P	b/b_P	Paired change in b/a	Paired change in b/b_P
Gamma 3.6 / 1.6 (primary)	1748	0.0150 [0.0101, 0.0206]	0.560 [0.362, 0.769]	37.4 [18.4, 72.4]	0.446 [0.427, 0.465]	1.26 [0.81, 1.73]	reference	reference
Gamma 3.2 / 1.42	1748	0.0119 [0.0080, 0.0164]	0.448 [0.292, 0.614]	37.6 [18.7, 72.1]	0.354 [0.340, 0.368]	1.27 [0.83, 1.74]	0.17 [-0.65, 0.89]	0.010 [-0.005, 0.028]
Gamma 3.2 / 2.1	1748	0.0111 [0.0074, 0.0151]	0.420 [0.279, 0.570]	37.9 [19.5, 71.3]	0.340 [0.322, 0.360]	1.23 [0.83, 1.68]	0.49 [-2.55, 3.04]	-0.023 [-0.082, 0.036]

State-week bootstrap with 2000 draws; the same weights are reused for every generation interval (paired), so the last two columns are paired differences from the primary specification. Gamma 3.2 / 1.42 keeps the primary coefficient of variation; 3.2 / 2.1 is the intrinsic generation-time mean and SD estimated from US household data [33].

Table S14 Held-out comparison of the affine and power loss models with paired uncertainty

Held-out unit	Folds	Mean power - affine score	Paired fold SE	z	Folds favouring affine	Bootstrap 95% CI	Bootstrap share favouring affine
State	51	-0.0020	0.0068	-0.30	26/51	[-0.0290, 0.0298]	0.43
Season	4	0.0046	0.0087	0.53	3/4	[-0.0295, 0.0655]	0.54

Score: weighted mean of $L/\mu + \log \mu$ on held-out rows; positive differences favour the affine model. Paired fold SE from the between-fold variation of the paired differences; state-week bootstrap with 1000 draws refits both models in every fold.

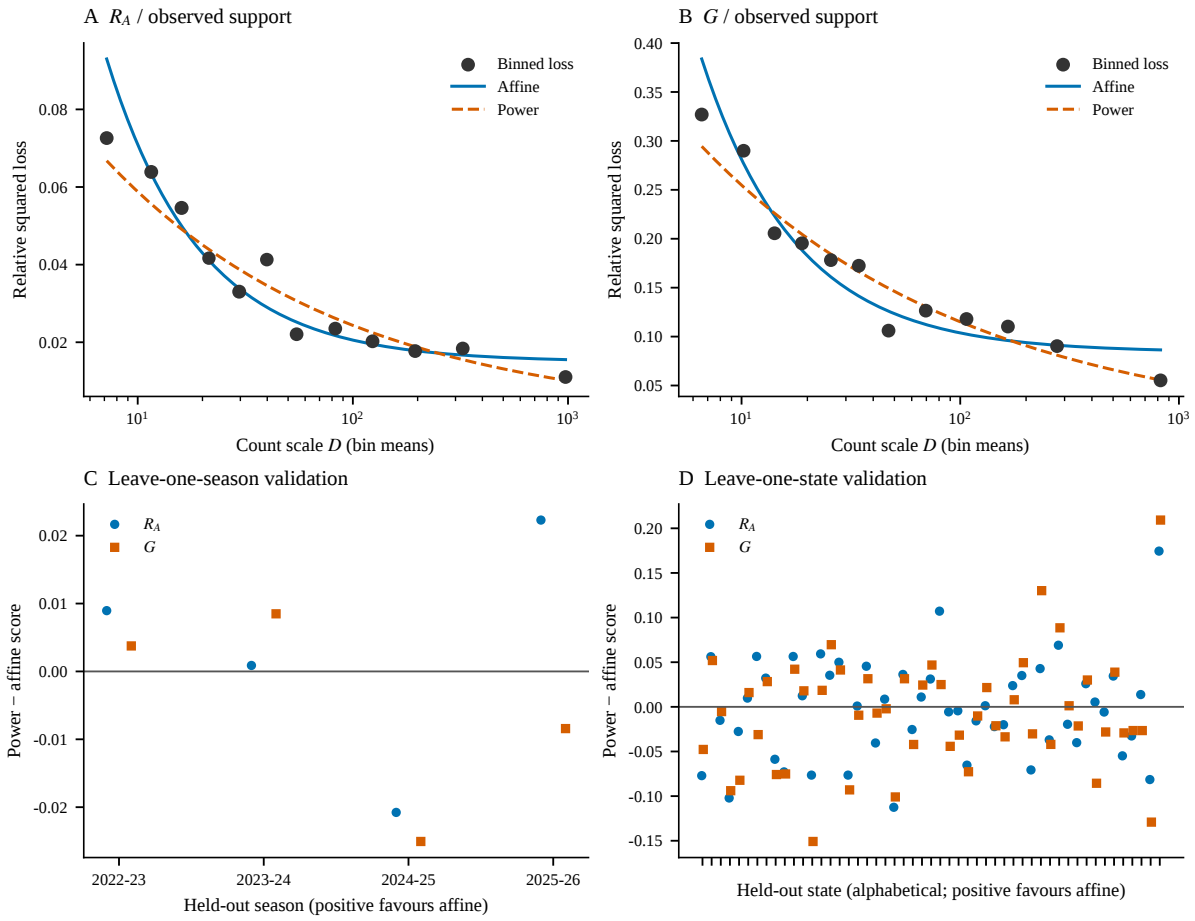


Figure S2 Affine and power-law forms of the dispersion curve. (A, B) Relative squared deviation averaged within count bins for R_A and G, with the fitted affine and power-law curves; bins are for display only. (C) Held-out power-minus-affine score difference for each left-out season. (D) The same for each left-out state. Positive differences favour the affine model.

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